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Roll No. : 241801304 Sub : Computer Networks

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completed

Exp.no: 1
Date: 16/7/26

Study of various network commands used in Linux and Windows

Windows commands:

1) arp (address resolution protocol)
2) hostname
3) ipconfig
4) nbtstat
5) netstat
6) nslookup
7) pathping
8) ping (Packet internet groupby)
9) route

Command Prompt - nslookup

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\TCS>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a               Displays current ARP entries by interrogating the current
                   protocol data.  If inet_addr is specified, the IP and Physical
                   addresses for only the specified computer are displayed.  If
                   more than one network interface uses ARP, entries for each ARP
                   table are displayed.
  -g               Same as -a.
  -v               Displays current ARP entries in verbose mode.  All invalid
                   entries and entries on the loop-back interface will be shown.
  inet_addr        Specifies an internet address.
  -N if_addr       Displays the ARP entries for the network interface specified
                   by if_addr.
  -d               Deletes the host specified by inet_addr. inet_addr may be
                   wildcarded with * to delete all hosts.
  -s               Adds the host and associates the Internet address inet_addr
                   with the Physical address eth_addr.  The Physical address is
                   given as 6 hexadecimal bytes separated by hyphens. The entry
                   is permanent.
  eth_addr         Specifies a physical address.
  if_addr          If present, this specifies the Internet address of the
                   interface whose address translation table should be modified.
                   If not present, the first applicable interface will be used.
Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.

C:\Users\TCS>hostname
DESKTOP-F4RMSER
```

```
C:\Users\TCS>hostname
DESKTOP-T4RMSER

C:\Users\TCS>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::d51f:7aad:f478:5629%7
   IPv4 Address. . . . . . . . . . . : 172.16.11.192
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1

C:\Users\TCS>nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

```
Command Prompt - nslookup

C:\Users\TCS>netstat

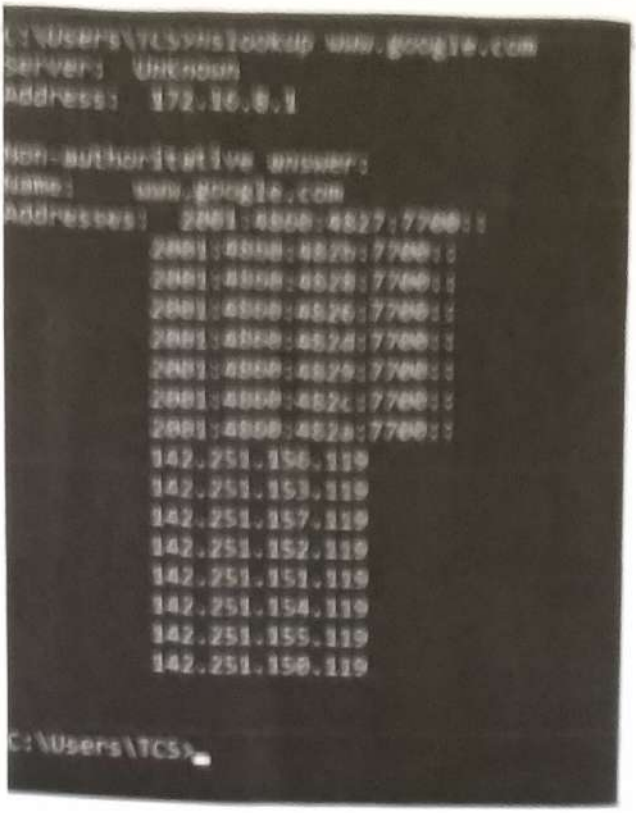
Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    172.16.11.192:51810    4.213.25.240:https     ESTABLISHED
  TCP    172.16.11.192:51812    a23-202-229-23:https   CLOSE_WAIT
  TCP    172.16.11.192:51813    a23-202-229-23:https   CLOSE_WAIT
  TCP    172.16.11.192:51814    a23-211-60-142:https   CLOSE_WAIT
  TCP    172.16.11.192:51815    a23-211-60-142:https   CLOSE_WAIT
  TCP    172.16.11.192:51816    a23-211-60-142:https   CLOSE_WAIT
  TCP    172.16.11.192:51817    a23-202-229-23:https   CLOSE_WAIT
  TCP    172.16.11.192:51818    a23-202-229-23:https   CLOSE_WAIT
  TCP    172.16.11.192:51819    a23-202-229-23:https   CLOSE_WAIT
  TCP    172.16.11.192:51820    a23-199-67-57:https    CLOSE_WAIT
  TCP    172.16.11.192:51821    a23-199-67-57:https    CLOSE_WAIT
  TCP    172.16.11.192:52699    123:http               TIME_WAIT
  TCP    172.16.11.192:52700    123:http               TIME_WAIT
  TCP    172.16.11.192:52701    150.171.28.12:https    ESTABLISHED
  TCP    172.16.11.192:52702    a23-202-229-22:https   CLOSE_WAIT
  TCP    172.16.11.192:52703    4.207.44.66:https      TIME_WAIT
  TCP    172.16.11.192:52704    a23-202-229-22:https   ESTABLISHED
  TCP    172.16.11.192:52705    4.213.25.242:https     ESTABLISHED
  TCP    172.16.11.192:52706    123:http               ESTABLISHED

C:\Users\TCS>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list      Loose source route along host-list.
    -h maximum_hops   Maximum number of hops to search for target.
    -i address        Use the specified source address.
    -n                Do not resolve addresses to hostnames.
    -p period         Wait period milliseconds between pings.
    -q num_queries    Number of queries per hop.
    -w timeout        Wait timeout milliseconds for each reply.
    -4                Force using IPv4.
    -6                Force using IPv6.
```



```
C:\Users\TCS>nslookup www.google.com
Server:  UnKnown
Address:  172.16.8.1

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:4827:7700::
          2001:4860:4827:7700::
          2001:4860:4828:7700::
          2001:4860:4826:7700::
          2001:4860:4824:7700::
          2001:4860:4829:7700::
          2001:4860:482c:7700::
          2001:4860:482a:7700::
          142.251.156.119
          142.251.153.119
          142.251.157.119
          142.251.152.119
          142.251.151.119
          142.251.154.119
          142.251.155.119
          142.251.150.119

C:\Users\TCS>
```

```
C:\Users\TCS>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t                 Ping the specified host until stopped.
                       To see statistics and continue - type Control-Break;
                       To stop - type Control-C.
    -a                 Resolve addresses to hostnames.
    -n count           Number of echo requests to send.
    -l size            Send buffer size.
    -f                 Set Don't Fragment flag in packet (IPv4-only).
    -i TTL             Time To Live.
    -v TOS             Type Of Service (IPv4-only. This setting has been deprecated
                       and has no effect on the type of service field in the IP
                       Header).
    -r count           Record route for count hops (IPv4-only).
    -s count           Timestamp for count hops (IPv4-only).
    -j host-list       Loose source route along host-list (IPv4-only).
    -k host-list       Strict source route along host-list (IPv4-only).
    -w timeout         Timeout in milliseconds to wait for each reply.
    -R                 Use routing header to test reverse route also (IPv6-only).
                       Per RFC 5095 the use of this routing header has been
                       deprecated. Some systems may drop echo requests if
                       this header is used.
    -S srcaddr         Source address to use.
    -c compartment     Routing compartment identifier.
    -p                 Ping a Hyper-V Network Virtualization provider address.
    -4                 Force using IPv4.
    -6                 Force using IPv6.
```

Command Prompt - nslookup

```
C:\Users\TCS>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

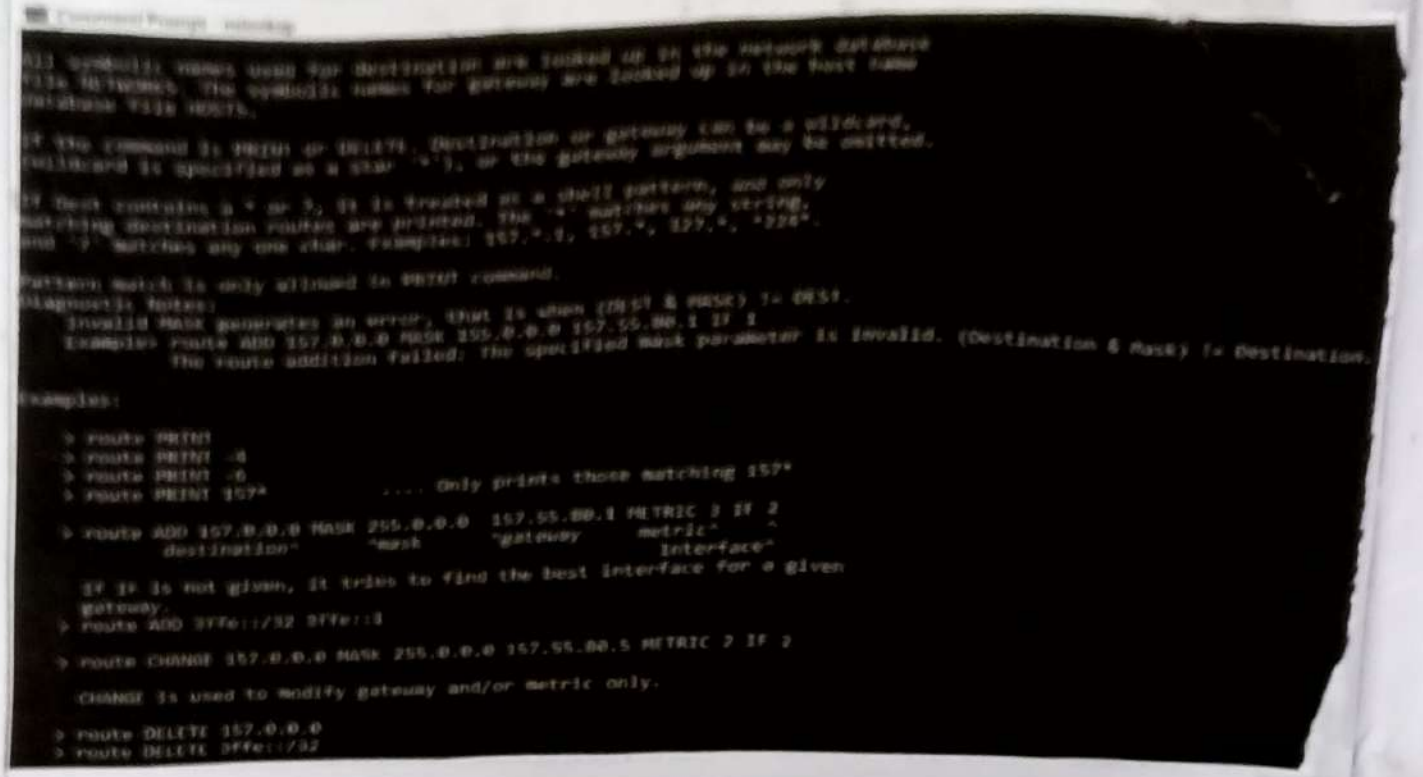
  -f               Clears the routing tables of all gateway entries.  If this is
                   used in conjunction with one of the commands, the tables are
                   cleared prior to running the command.

  -p               When used with the ADD command, makes a route persistent across
                   boots of the system. By default, routes are not preserved
                   when the system is restarted. Ignored for all other commands,
                   which always affect the appropriate persistent routes.

  -4               Force using IPv4.

  -6               Force using IPv6.

  command          One of these:
                     PRINT     Prints  a route
                     ADD       Adds    a route
                     DELETE    Deletes a route
                     CHANGE    Modifies an existing route
  destination      Specifies the host.
  MASK             Specifies that the next parameter is the 'netmask' value.
  netmask          Specifies a subnet mask value for this route entry.
                   If not specified, it defaults to 255.255.255.255.
  gateway          Specifies gateway.
  interface        the interface number for the specified route.
  METRIC           specifies the metric, ie. cost for the destination.
```

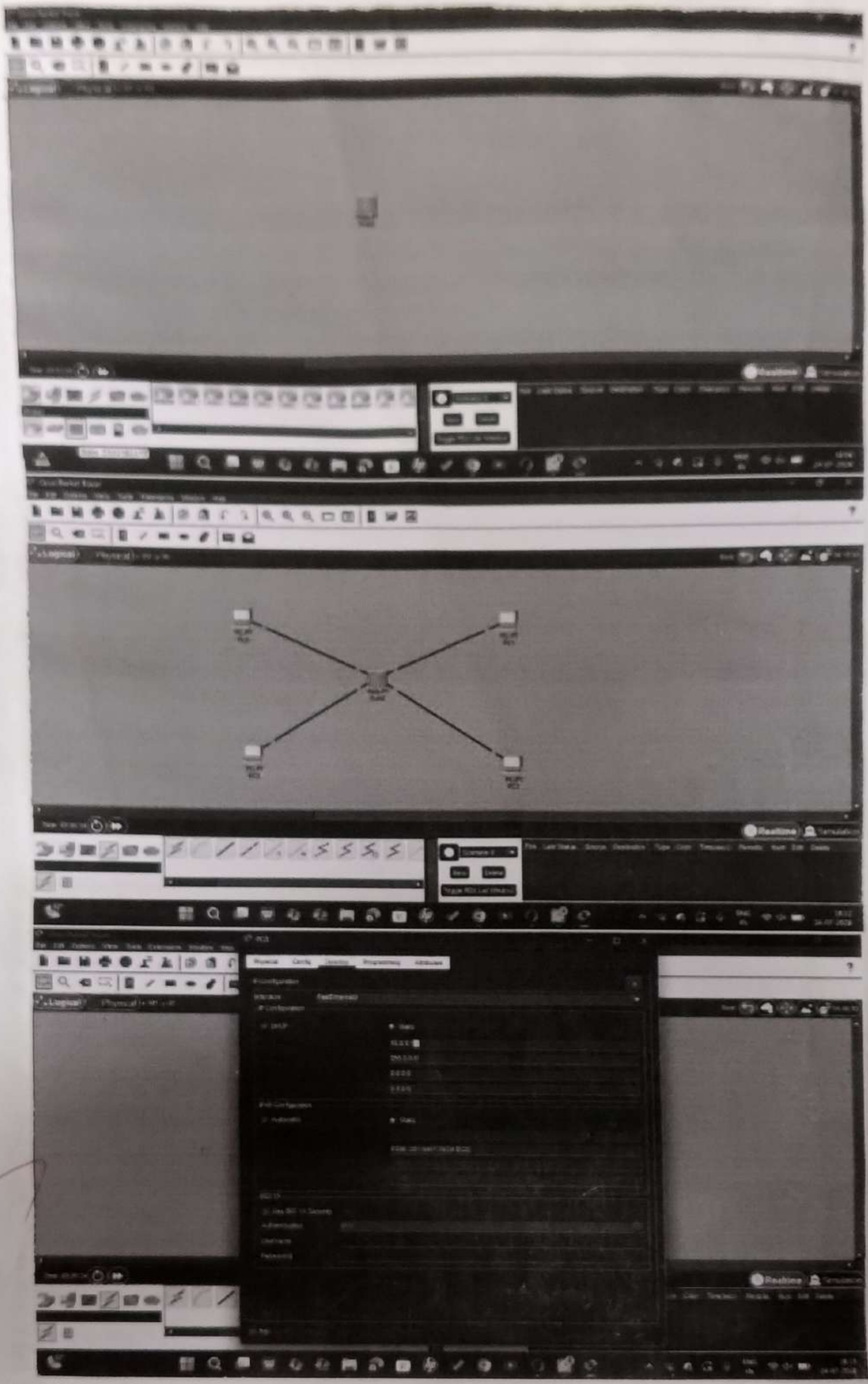


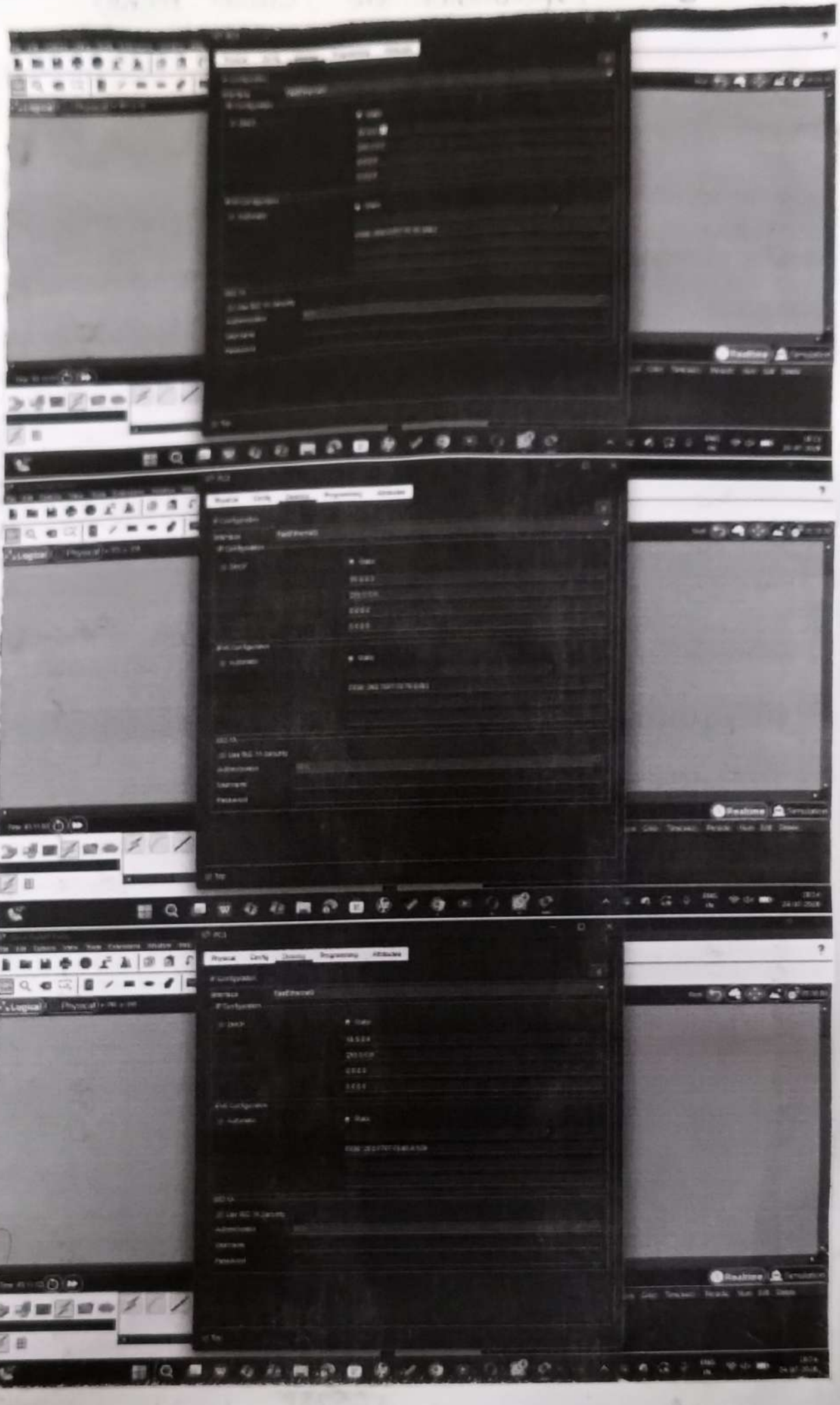
Result:

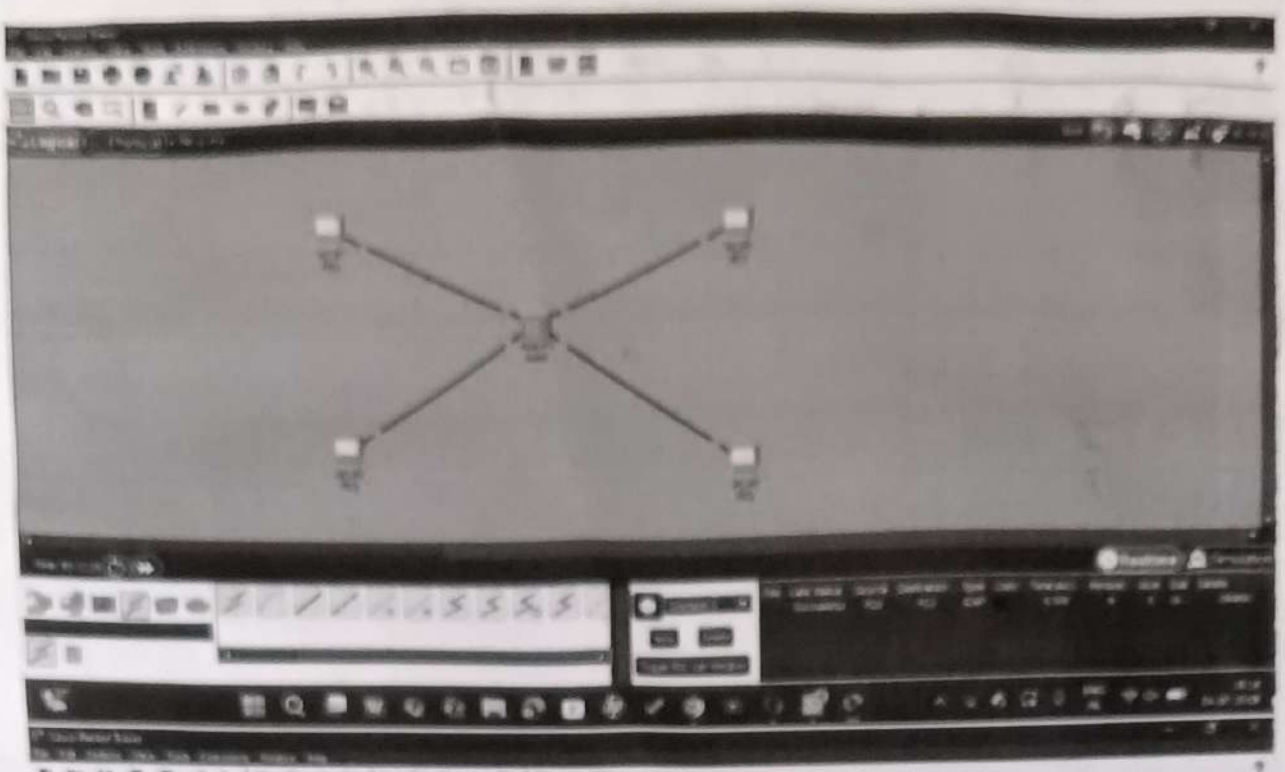
Thus, various network commands used in Linux and Windows has been verified and executed successfully.

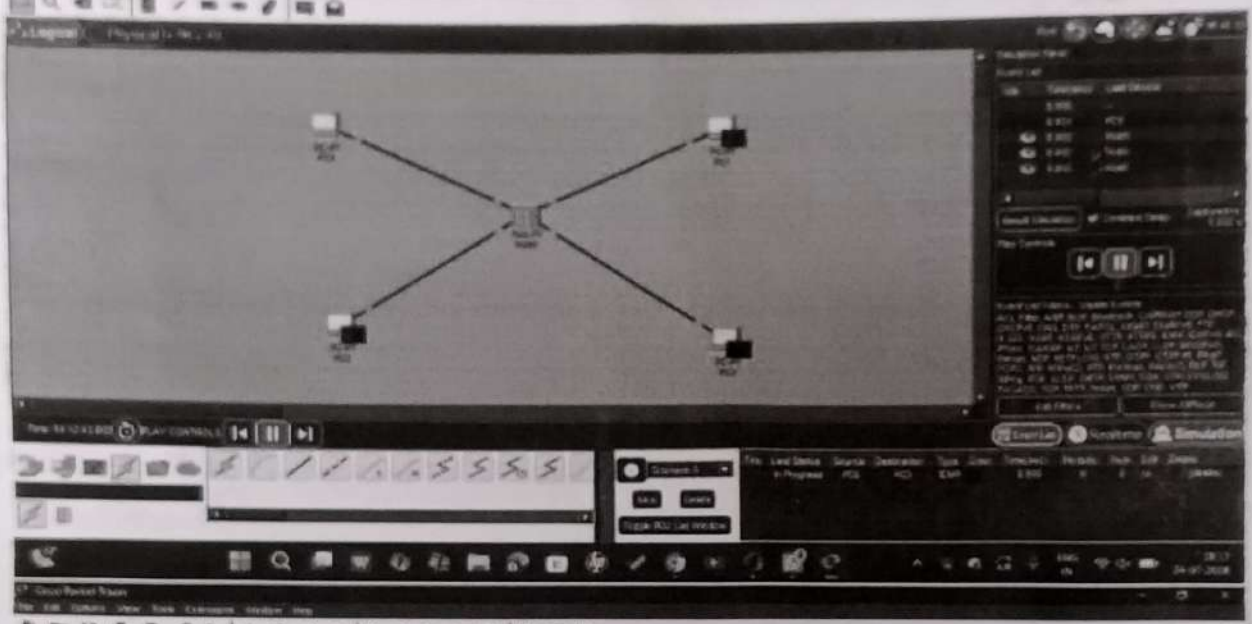
Exp. no: 3 Experiments on CISCO PACKET TRACKER (Simulation Tool)

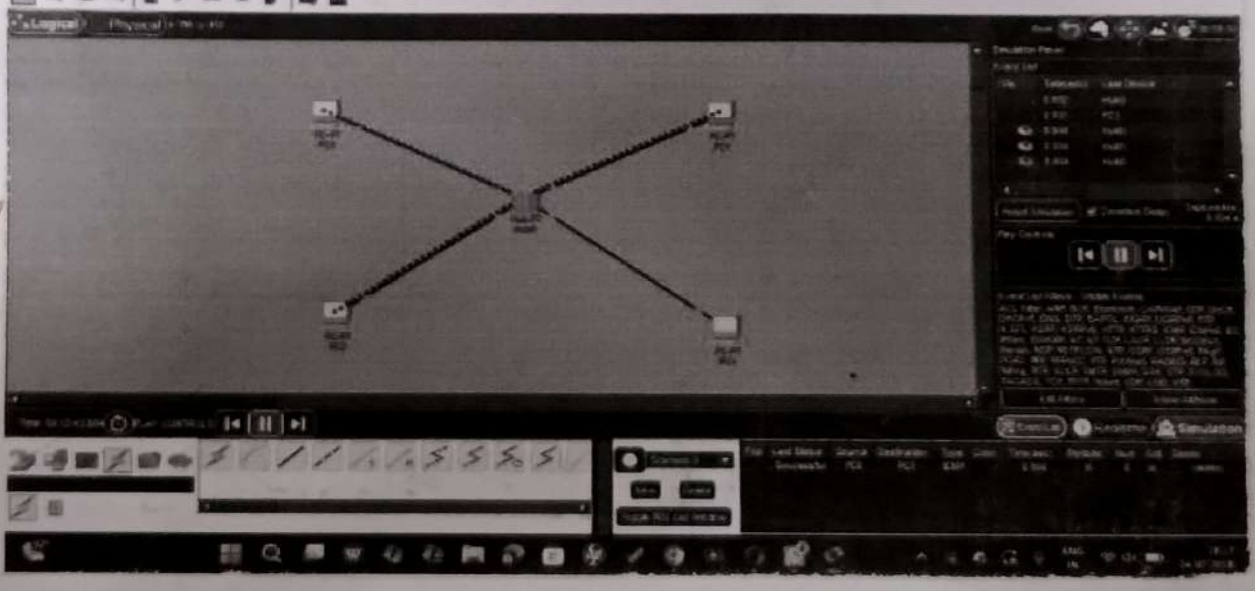
Date: 22.7.26

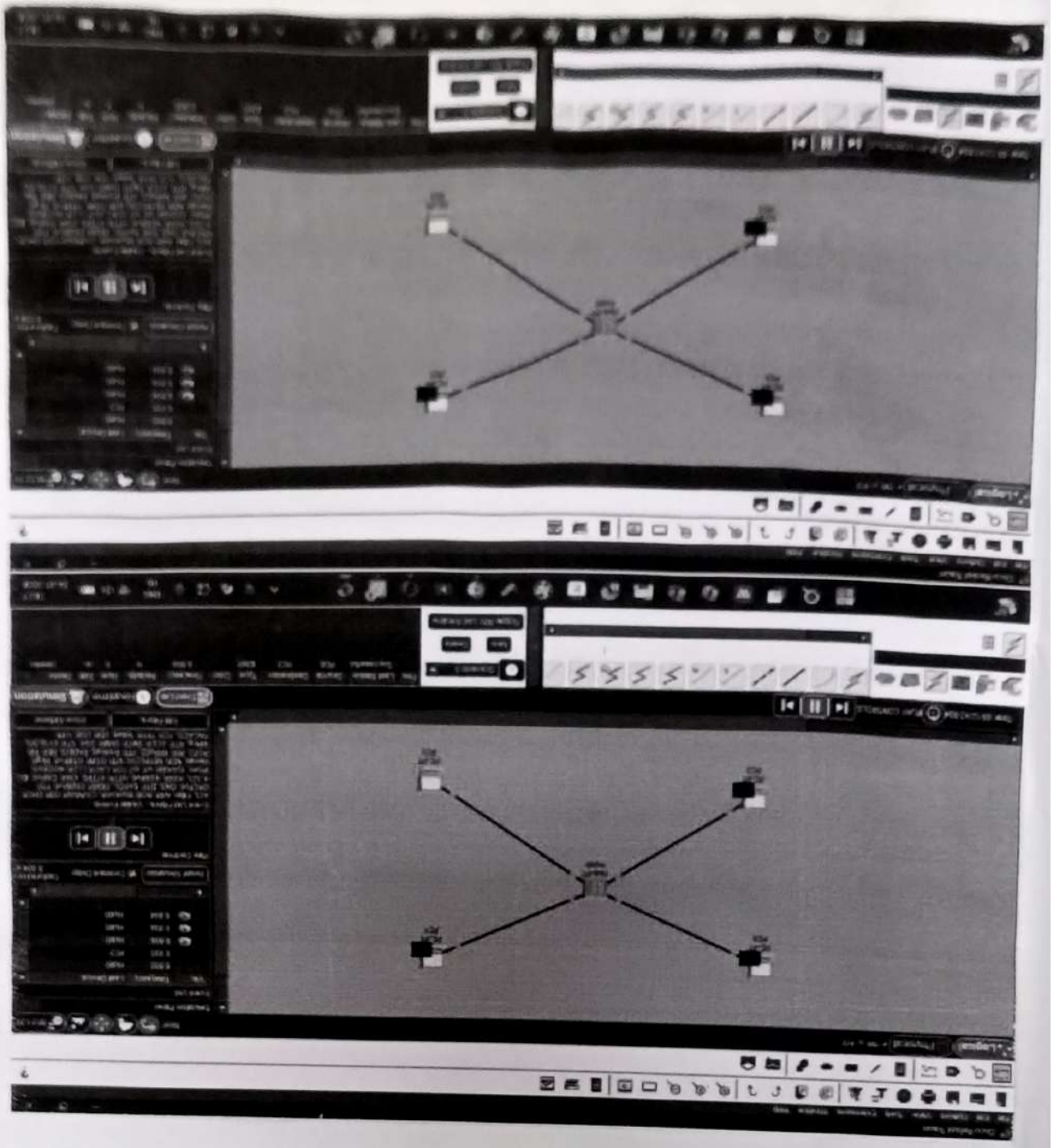












Explanation:

1. Opened Cisco Packet Tracer and created a new workspace.
2. Added one Hub and four PCs to the workspace.
3. Connected all the PCs to the Hub using Copper Straight - Through cables.
4. Configured a unique IP address to each PC through Desktop → IP configuration
5. Ensured that all the PCs belonged to the same network for successful communication.
6. Switched to simulation mode to observe how data packets travel in the network.
7. Used the Add Simple PDU tool to send packets from one PC to another.
8. Observed the packet movement and verified that the data reached the destination successfully.
9. Confirmed that proper cabling and correct IP configuration enabled communication b/w all the devices.
10. Understood the basic working of a LAN and how data transmission occurs b/w connected computers.

Result:

A simple LAN was successfully created in Cisco Packet Tracer using one hub and four PCs. All devices were configured with valid IP addresses, and successful packet transmission b/w the PCs confirmed proper network connectivity and communication.

Exp.no: 2
Date: 22.7.24

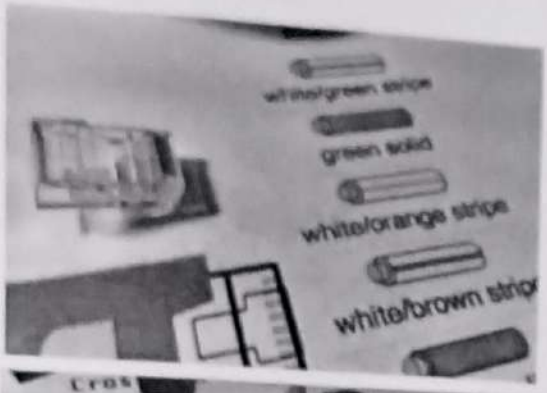
Study of different types of Network cables

a) Understand different types of network cable.

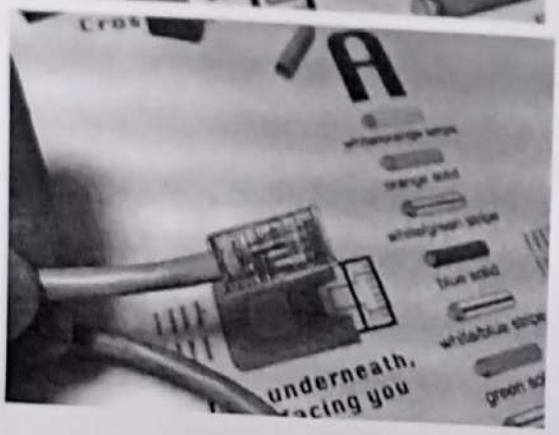
Different type of cables used in networking are:

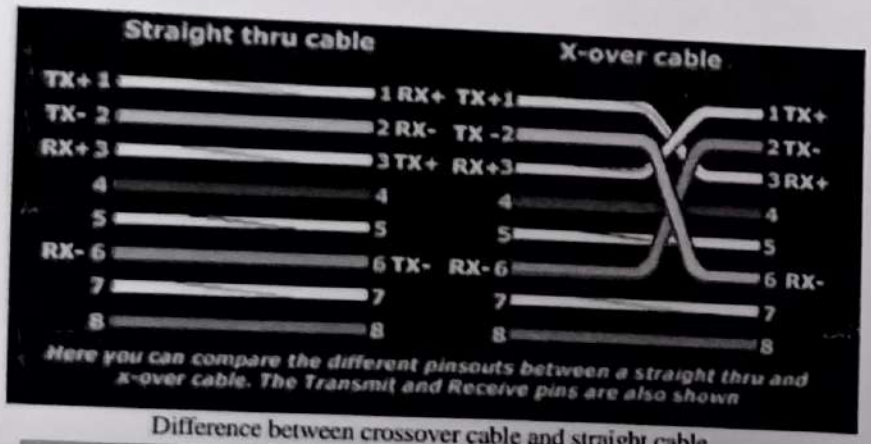
1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic cable

Cable Type	Category	Maximum Data Transmission	Advantages/ Disadvantages	Application/Use
UTP	Category 3	10 bps	Advantages: • Cheaper in cost. • Easy to install as they have a smaller overall diameter. Disadvantages: • More prone to (EMI) Electromagnetic interference and noise	10 Base-T Ethernet
	Category 5	Up to 100 Mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	1 Gbps		Fast Ethernet, Gigabit Ethernet
STP	Category 6, 6a	10 Gbps	Advantages: • Shielded • Faster than UTP. • Less susceptable to noise and interference Disadvantages: • Expensive • Greater installation effort.	Gigabit Ethernet, 10G Ethernet (55m) Widely used in data centres.
SSTP	Category 7	10 Gbps		Gigabit Ethernet, 10G Ethernet (100m)

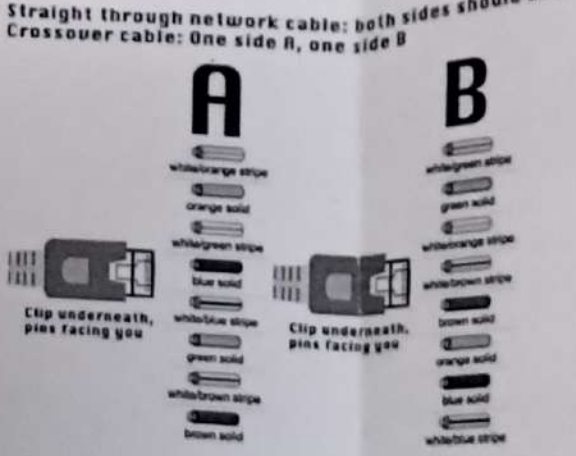








Difference between crossover cable and straight cable











Coaxial Cable	RG-6 RG-59 RG-11	10 - 100 Mbps	Advantages: • High Bandwidth • Immune to interference • Low loss bandwidth • Versatile Disadvantages: • Limited distance • Cost • Size is bulky	Speed of signal is 500 m. Television network High speed internet connections
fibre optics cable	Single mode Multi mode	100 Gbps	Advantages: • High speed • High bandwidth • High security • Long distance Disadvantages: • Expensive • Requires skilled installers.	• Maximum distance of fibre optics cable is around 100 meters.

b) Make your own Ethernet Cross-Over Cable/Straight Cable:

Tools and parts needed:

- Ethernet cabling, CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.
- A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
- Two RJ45 plugs.
- Optional two plug shields.

Step 1:
To start construction of the device, begin by threading shields onto the cable.

Step 2:
Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3:
After, you will need to untangle the wires; there should be four "twisted pairs". Referring back to the sheet, arrange them from top to bottom. One end should be in arrangement A and other in B.

Step 4:
Arrange the wires in the correct order, trim them evenly, and insert them into the RJ45 plug with the clip facing away and the gold pins facing you.

Step 5:
Push the cable fully into the plug so the cable jacket is inside the connector. Trim and reinsert if needed.

Step 6:
Place the RJ45 plug into the trimming tool and trim it firmly without applying excessive force.

Step 7: Repeat for the other end using B for a crossover cable or A for a straight-through cable. Test the cable by connecting two devices.

Result:

Thus, different types of networking cables have been studied successfully.

Logical
Realtime
Logical
Physical

Network Connections
Control Panel
Network and Internet
Network Connec
Ethernet Properties
Networking
Sharing
Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address:
10 . 1 . 1 . 1
Subnet mask:
255 . 0 . 0 . 0
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...

IP Configuration
IP Configuration
DHCP
Static
IP Address
Subnet Mask
Default Gateway
DNS Server
10.1.1.1
255.0.0.0
IP Configuration
IP Configuration
DHCP
Static
IP Address
Subnet Mask
Default Gateway
DNS Server
10.1.1.2
255.0.0.0

Ex.no: 4
Date: 30.7.26

Setup and Configure a LAN using a Switch and Ethernet cables

Aim:

To setup and configure a LAN (Local Area Network) using a Switch and Ethernet cables.

LAN:

A Local Area Network (LAN) refers to a network that connects devices within a limited area, such as an office building, school or home. It enables users to share resources, including data, printers and internet access. LAN connects devices to promote collaboration and transfer information between users, such as computers, printers, servers and switches. A LAN switch serves as the primary connecting device, managing and directing communications within the local network. Each connected device on a LAN switch can communicate directly with each other, allowing for fast and secure data transfer.

How to set up a LAN:

Step 1: Plan and design an appropriate network topology taking into account network requirements and equipment location.

Step 2: You can take 4 computers, a Switch with 8, 16, or 24 ports which is sufficient for networks of these sizes, and 4 Ethernet cables.

Step 3: Connect your computers to network switch via an Ethernet cable, which is as simple as plugging one end of the Ethernet cable into your computer and the other end into your network switch.

Step 4: Assign IP address to your PCs:

1. Log on to the client computer as Administrator or Owner.
2. Click Network and Internet Connections.

3. Right click Local Area Network connection/ Ethernet → go to properties → Select Internet Protocol (TCP/IPv4) → Click on Properties → Select use the following IP address option and assign ip address.

Similarly assign IP address to all PCs connected to switch. PC1 - IP address: 10.1.1.1

PC2 - IP address: 10.1.1.2

Step 5: Configure a network switch:

1. Connect your computer to the switch: To access the switch's web interface, you will need to connect your computer to the switch using an Ethernet cable.

2. Log in to the web interface: Open a web browser and enter the IP address of the switch in address bar. This should bring up the login page for the switch's web interface. Enter the username and password to log in.

3. Configure basic settings: Once you're logged in, you will be able to configure basic settings for the switch.

4 Assign IP address as: 10.1.1.5, subnet mask 255.0.0.0.

Step 6: Check the connectivity b/w switch and other machine by using ping command in the command prompt of the device.

Step 7: Select a folder → go to properties → Click Sharing tab → share it with everyone on the same LAN.

Step 8: Try to access the shared folder from others computers of the network.

Result:

The LAN was successfully set up using a switch and Ethernet cables. IP addresses where assigned to all computers, connectivity was verified using the ping command, and a folder was successfully shared and accessed b/w computers on the same network.

Ex.no: 5
Date : 06.08.2025

Packet Capture Using Wireshark

Aim:

To capture filter inspect and analyze network packets using the wireshark packet analyzer.

Algorithm:

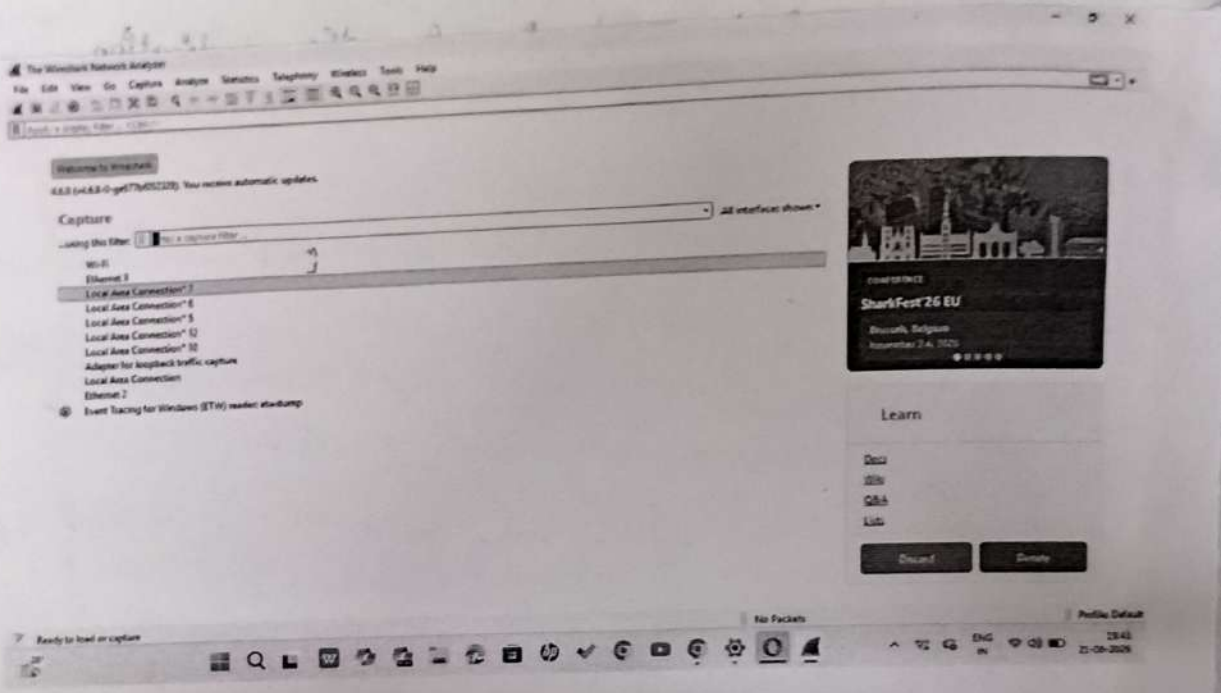
1. Open Wireshark and launch the application.
2. Select the active network interface such as Wi-fi or Ethernet.
3. Go to Capture → Options and configure the capture settings.
4. Set the capture limit and start the packet capture and generate network traffic.
5. Observe the captured packets in the packet list.
6. Apply display filters such as TCP, UDP, ARP and DNS.
7. Select packets to inspect their detailed protocol information and flow graph.
8. Stop the capture after 100 packets and save the captured file for future analysis.

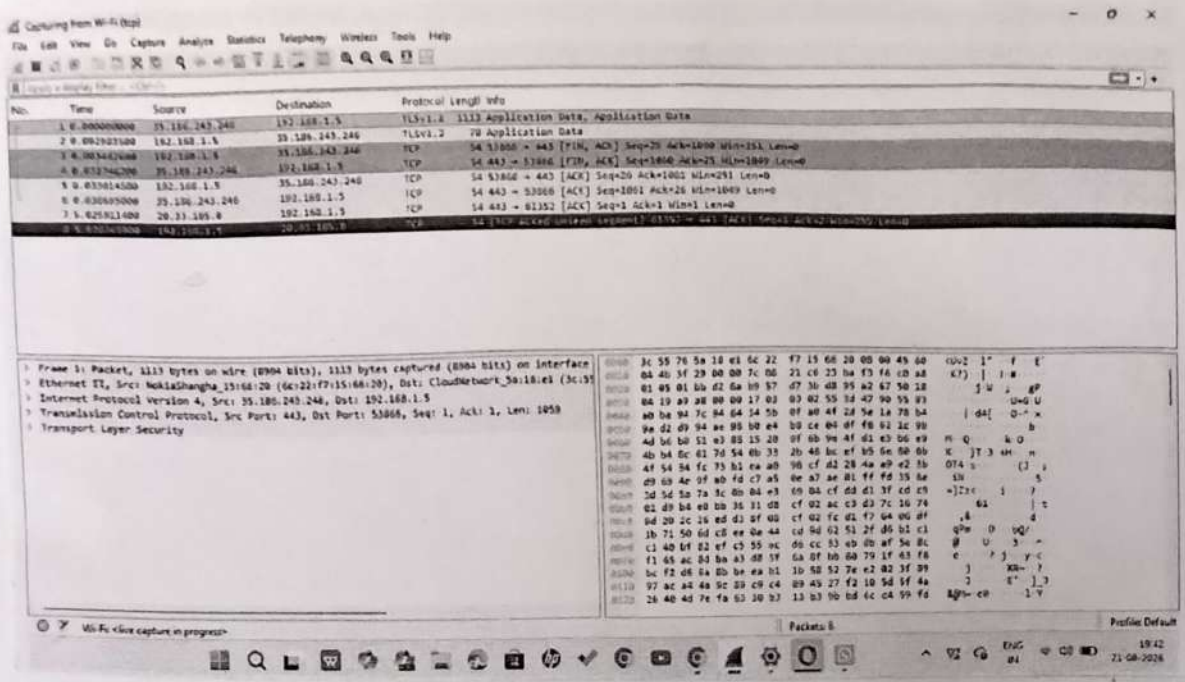
Procedure:

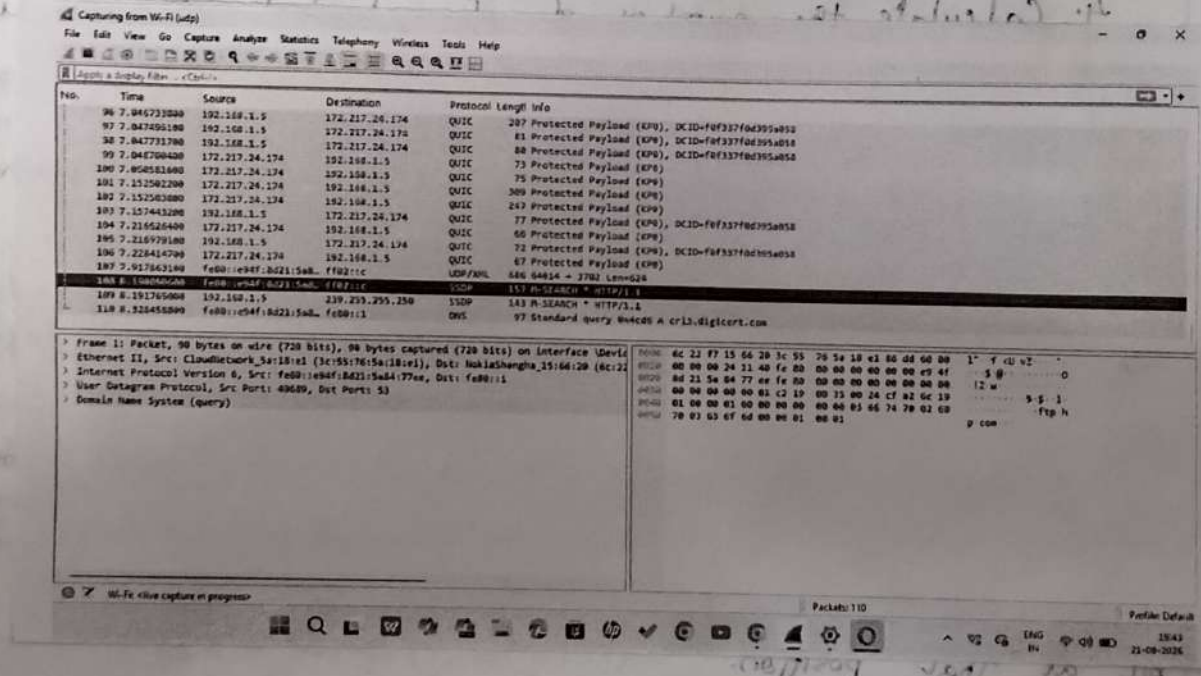
* Launch wireshark and subset of active network interface.
* Go to capture → options & set up capture filter after 100 packets.
* Click start to capture packet
  * TCP/UDP : tcp || udp

- ARP : arp
- DNS : dns
- HTTP : http

* Save the capture file to future analysis.







Result:

Successfully captured 100 network packets, applied display filters to TCP, UDP analyzed Packet details using wireshark.

Ex. no: 6 Implementation of Error Detection and
Date: 13.8.26 correction using HAMMING Code

Aim:

To write a program to implement error detection and correction using Hamming code concept. Make a test run to input data stream and verify error correction feature.

Algorithm:

1. Start
2. Read the text data from the sender.
3. Convert each character of the text into its 8-bit ASCII binary form.
4. Calculate the number of of redundant bits required using Hamming Code.
5. Insert redundant bits at position 1, 2, 4, 8....
6. Calculate the parity bits and generate Hamming Code.
7. Store the generated code in the chanel file.
8. At the receiver, read the data from the chanel file.
9. Calculate the parity bits again to find the error position.
10. If the error position is non-zero, correct the bit at that position.
11. Remove the redundant bits.
12. Convert the remaining binary data into ASCII characters.
13. Display the corrected message.
14. Stop.

Code:

```
def hamming(d):
    m = len(d) ; r=0
    while 2**r < m+r+1 :r+=1
    a=['0'] *(m+r) ; j=0
    for i in range(1, len(a)+1):
        if i&(i-1) : a[i-1] = d[j]; j+=1
    for p in [1<<i for i in range(r)]:
        x=0
        for i in range (1, len(a)+1):
            if i&p : x^= int(a[i-1])
        a[p-1] = str(x)
    return ''.join(a)
s=input("Enter data: ")
b=''.join(format(ord(c),'08b') for c in s)
c= hamming(b)
open("channel","w").write(c)
print("Binary data :", b)
print("Channel data :", c)

c= open("channel").read()
e= int(input("Enter error position (0 for no error): "))
if e:c = c[:e-1] + str(1-int(c[e-1])) + c[e:]
r=0
while 2**r < len(c)+1 : r+=1
x=0
for p in [1<<i for i in range(r)]:
    z=0
    for i in range(1, len(c)+1):
        if i&p : z^= int(c[i-1])
    if z: x+=p
print("Error position :", x)
if x: c = c[:x-1] + str(1-int(c[x-1])) + c[x:]
d=''.join(c[i] for i in range(len(c)) if (i+1)&i)
print("Received Data :", ''.join(chr(int(d[i:i+8],2)) for i in range(0, len(d), 8)))
```

Output:

Enter data: A

Binary data: 01000001

Channel data: 110100001011

Enter error position (0 for no error): 5

Error position: 5

Received Data : A

Result:

Hamming code successfully detects the error position and corrects the erroneous bit, recovering the original data.

Ex.no: 7 Flow Control at the Data Link
Date: 20.8.26 Layer Using the Sliding Window Protocol

Aim:

To Implement Flow Control at the Data Link Layer using the Sliding Window Protocol and simulate the transmission of frames from one node to another.

Algorithm:

1. Start the program.
2. Set the total number of frames and window size.
3. Sender sends frames equal to the current window size.
4. Receiver receives the frames and sends acknowlegements.
5. After receiving ACKs, the sender slides the window forward.
6. Continue untill all frames are transmitted.
7. Display the frame transmission and acknoladgement process.
8. Stop.

Code:

```
def sliding_window(total_frames, window_size):
    sender = 0
    print("Sliding Window Protocol")
    print("----------")
    while sender < total_frames:
        end = min(sender + window_size, total_frames)
        print("\n Sender sends frames:")
        for frame in range(sender, end):
            print("Frame", frame, "sent")
```

```
        print("Receiver sends acknoledgement:")
        for frame in range(sender, end):
            print("ACK", frame, "received")

        sender = end
        if sender < total_frames:
            print("Windows slides forward..")
    print("In All frames transmitted successfully.")
total_frames = int(input("Enter total number of frames:"))
window_size = int(input(total_frames, window_size)
sliding_window(total_frames, window_size)
```

Output:

```
Enter total number of frames: 8
Enter window size = 3
Sliding Window Protocol

Sender sends frames:
Frame 0 sent
Frame 1 sent
Frame 2 sent

Receiver sends acknowledgements:
ACK 0 received
ACK 1 received
ACK 2 received
Window slides forward..

Sender sends frames:
Frame 3 sent
Frame 4 sent
Frame 5 sent

Receiver sends acknowledgements:
ACK 3 received
```

Ack 4 received
Ack 5 received
Window slides forward...

Sender sends frames
Frame 6 sent
Frame 7 sent

Receiver sends acknowledgements
Ack 6 received
Ack 7 received

All frames transmitted successfully.

Results

The Sliding Window Protocol was successfully implemented in Python. The program simulated the transmission of frames between a sender and receiver using a fixed-size window, with acknowledgments causing the window to slide forward.

Ex.no: 8a
Date:

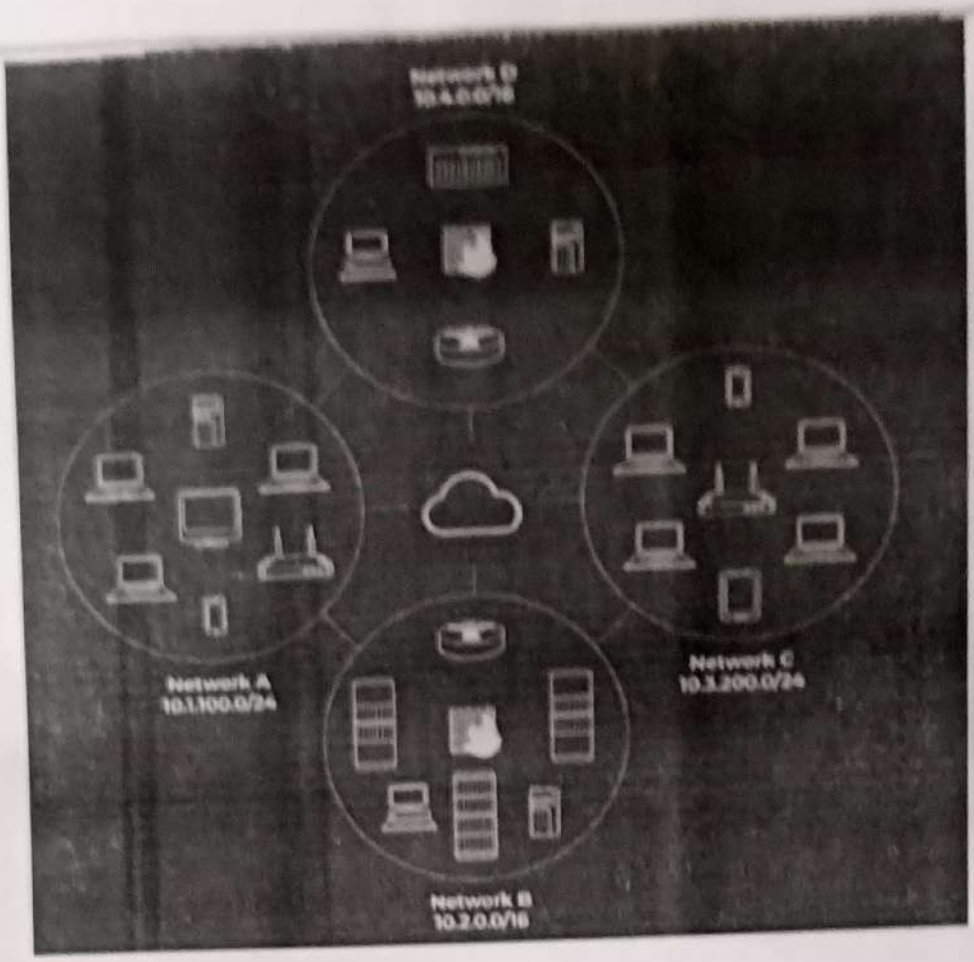
Nmap Live Host Discovery Using ARP, ICMP, TCP and UDP

Aim:

To discover live hosts using Nmap scans such as ARP, ICMP, TCP and UDP on the TryHackMe platform.

Algorithm:

1. Open the TryHackMe platform and access the Nmap Live Host Discovery room.
2. Study the concept of live host discovery before performing network scanning.
3. Start with an ARP scan to identify live hosts in the local network.
4. Observe the ARP requests and responses from the target hosts.
5. Perform a ICMP scan to check whether the target systems respond to ICMP requests.
6. Perform a TCP SYN ping scan by sending TCP packets to the required ports.
7. Perform a TCP ACK ping scan and observe the response from the target host.
8. Perform a UDP ping scan to determine whether the target system is reachable.
9. Compare the results obtained from ARP, ICMP, TCP and UDP host-discovery methods.
10. Record the live hosts identified by Nmap and study the scan results.



Send Packet

From:

computer1

To:

computer1

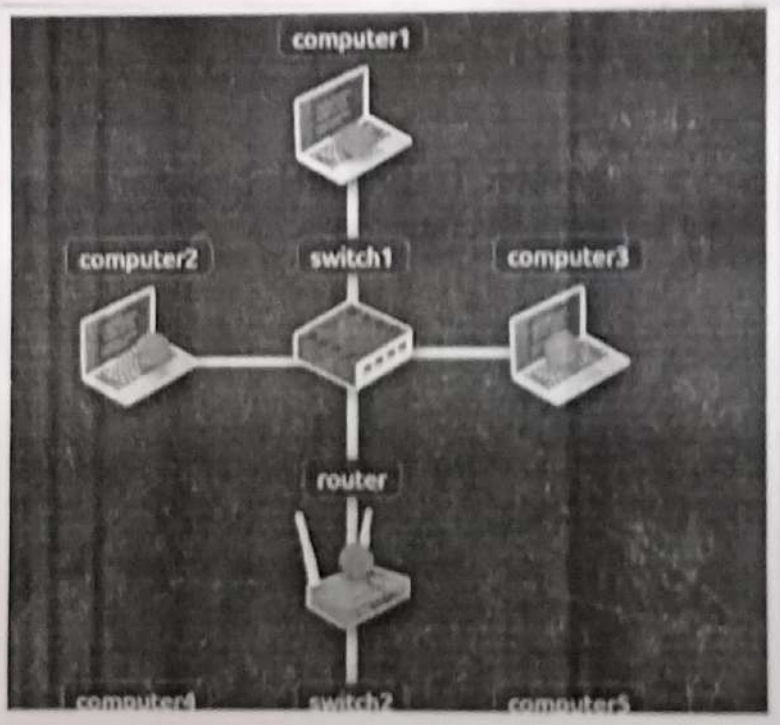
Packet Type:

arp_request

Data:

computer6

Send Packet



Send Packet

From:

computer4

To:

computer4

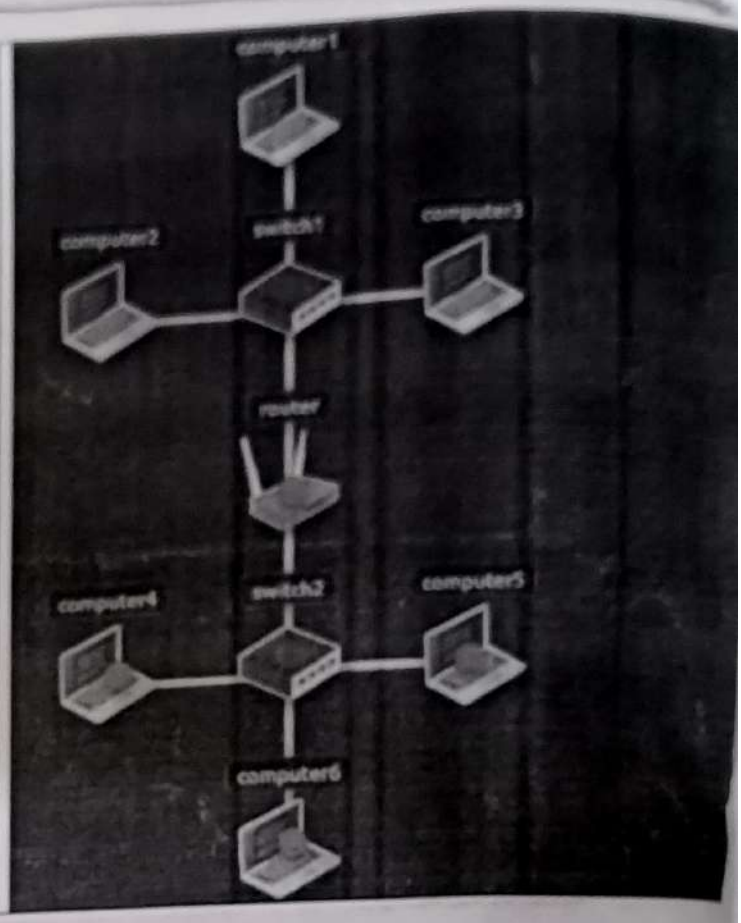
Packet Type:

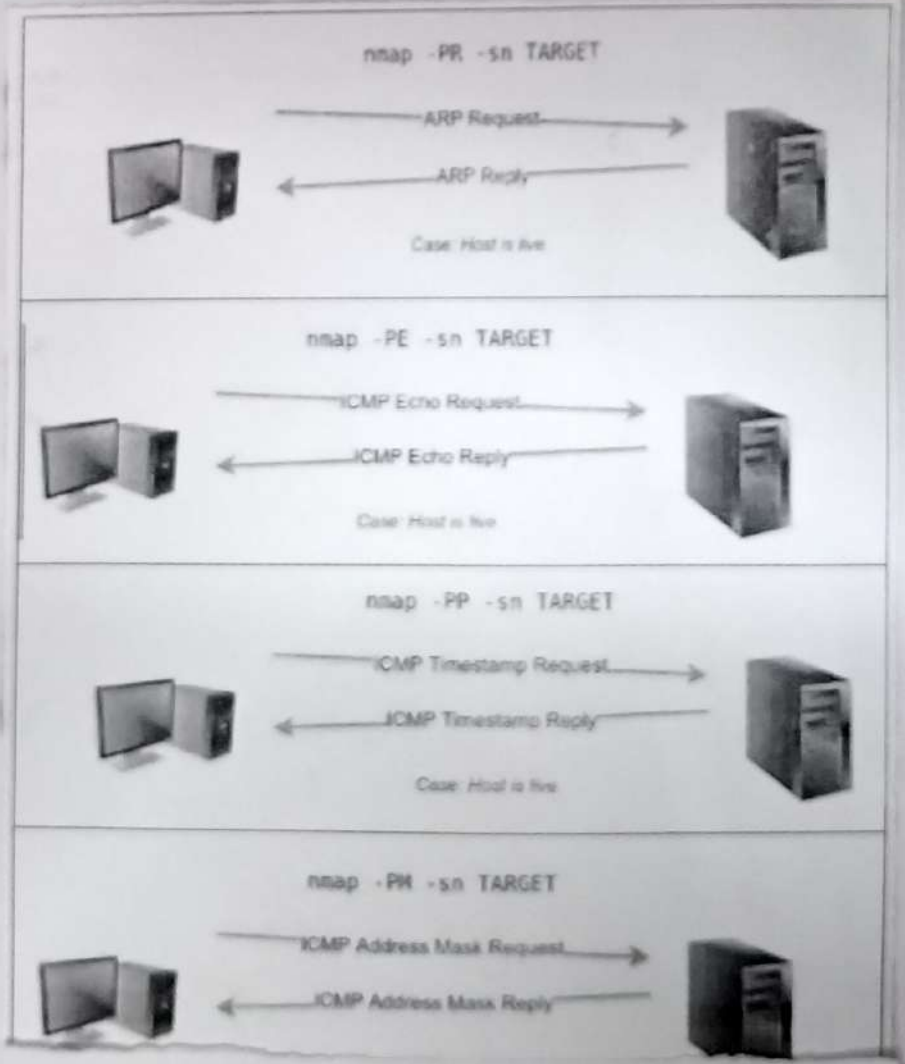
arp_request

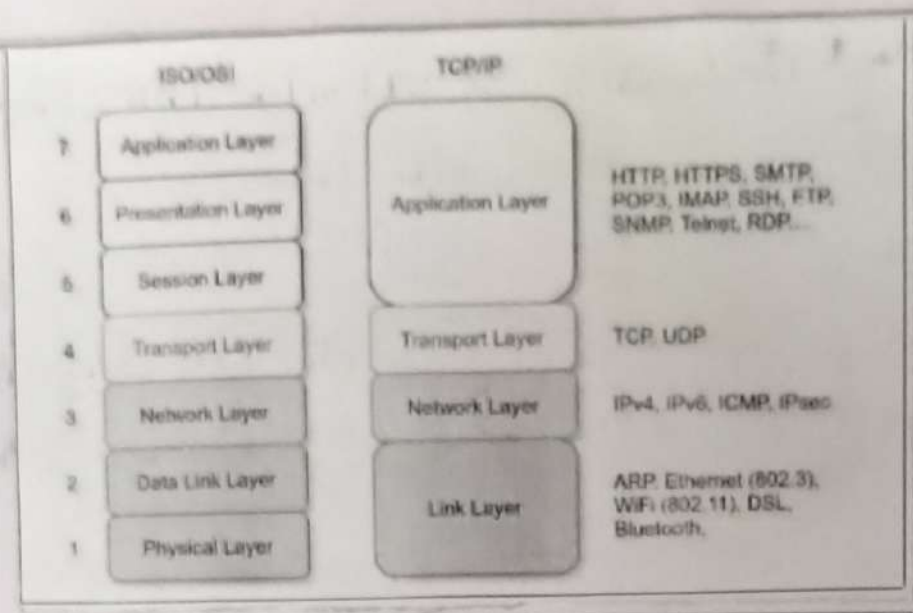
Data:

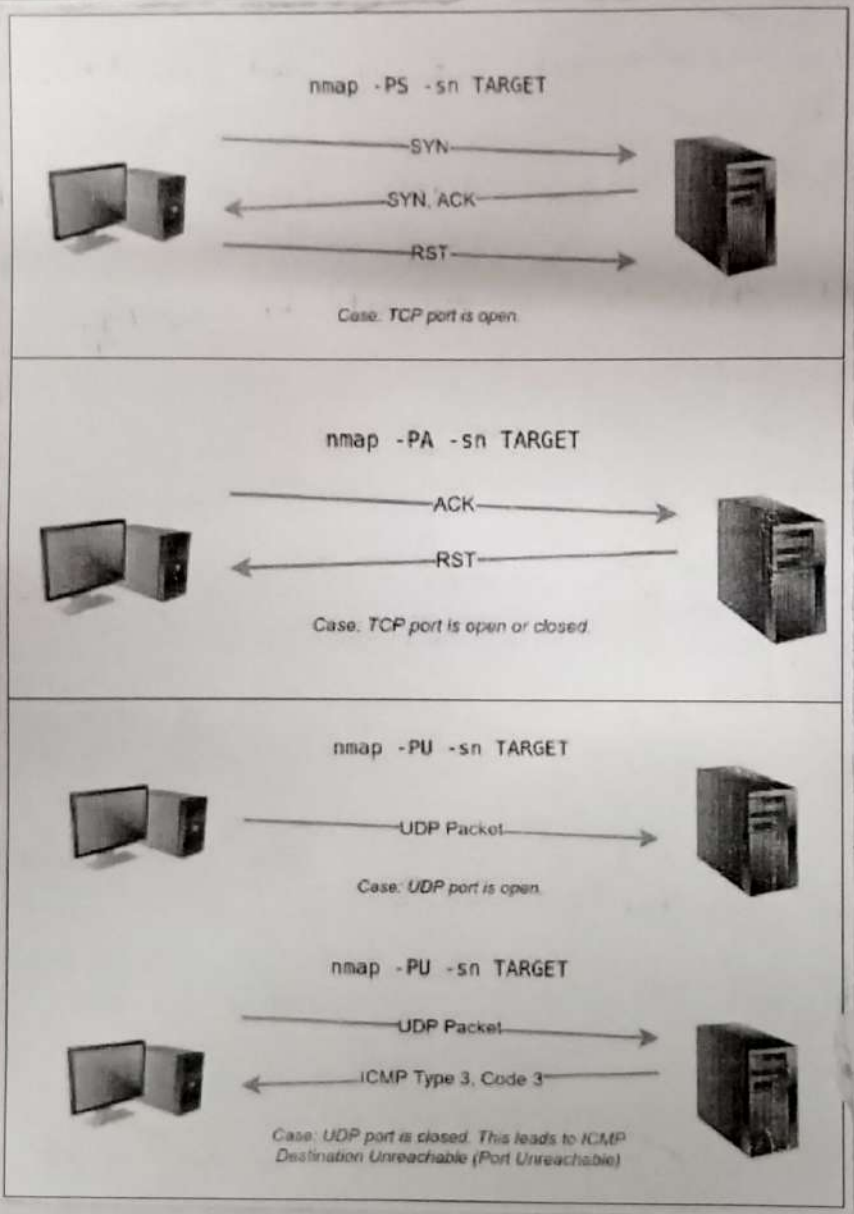
computer6

Send Packet









Result:

Thus, the live hosts were successfully discovered using ARP, ICMP, TCP and UDP scanning techniques in Nmap.

Ex.no: 8b
Date:

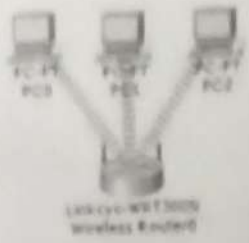
Configuration of Wireless LAN using Cisco Packet Tracer

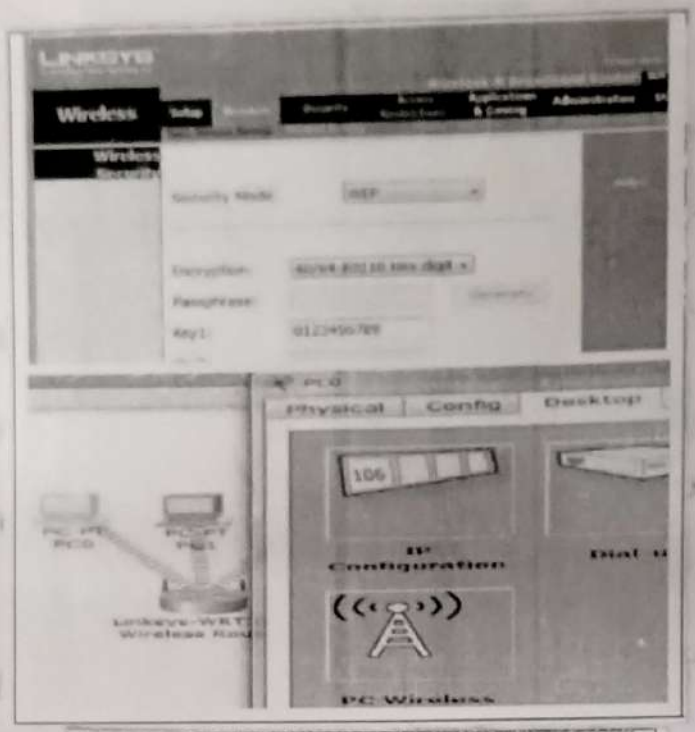
Aim:

To design and configure a Wireless LAN using Cisco Packet Tracer with three PCs connected to a Linksys Wireless Router.

Algorithm:

1. Open Cisco Packet Tracer and create a new workspace.
2. Select a Linksys Wireless Router and place it in the workspace.
3. Add three PCs to the network topology.
4. Configure the basic network settings of the wireless router.
5. Configure the wireless network SSID as MotherNetwork.
6. Configure the required WEP security for the wireless network.
7. Open the wireless configuration of the first PC and select MotherNetwork.
8. Enter the configured security key and connect the first PC to the router.
9. Repeat the wireless configuration for the second and third PCs.
10. Verify the wireless connections and test communication b/w the connected PCs.







Result:

Thus the Wireless LAN was successfully designed and configured using Cisco Packet Tracer with three PCs connected to the Linksys Wireless Router.

Ex.no: 9
Date :

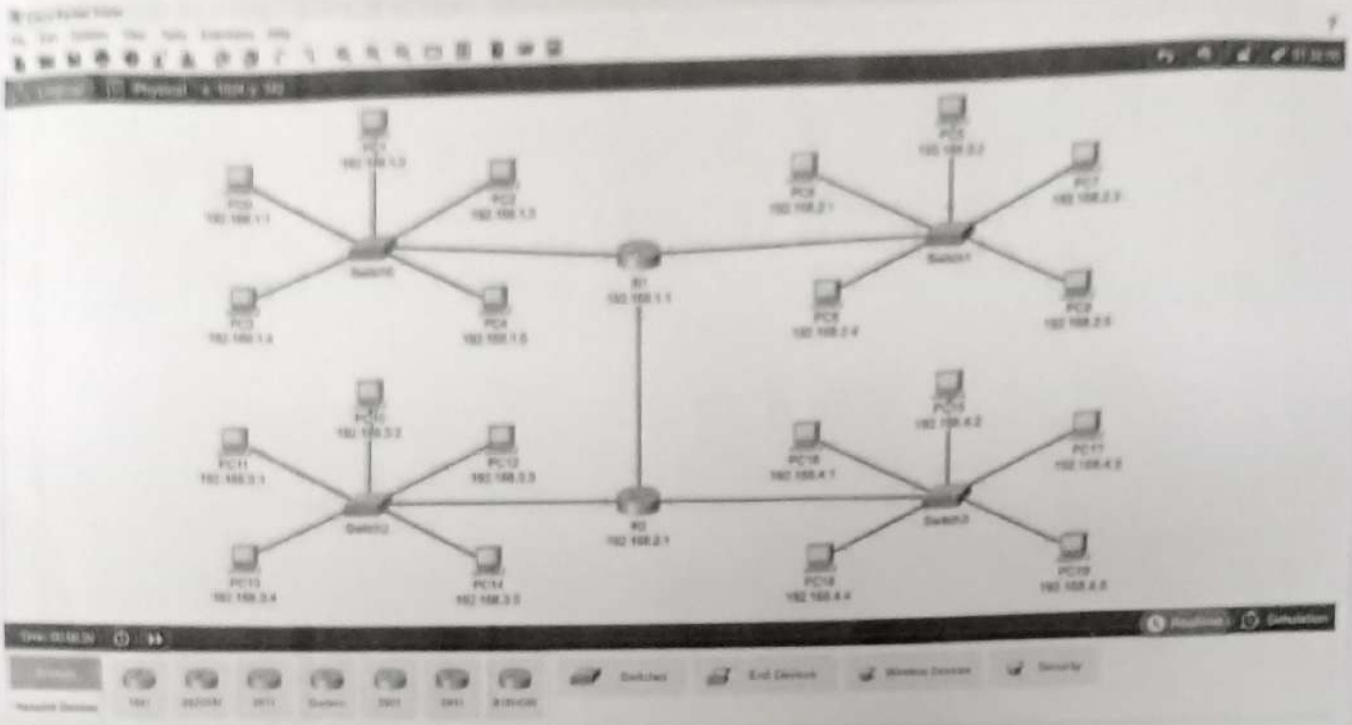
Subnetting

Aim:

To design and configure a subnetted network using Cisco Packet Tracer.

Algorithm:

1. Open Cisco Packet Tracer.
2. Add a router, switch and PCs.
3. Connect the devices using suitable cables.
4. Open the router CLI.
5. Configure the router interfaces with IP addresses and subnet masks.
6. Enable the interfaces using no shutdown.
7. Configure the IP address and subnet mask of each PC.
8. Set the appropriate default gateway for each PC.
9. Use the ping command to test connectivity.
10. Verify communication b/w the different subnets.



Result:

Thus, the subnetted network was successfully configured and connectivity was verified using ping.

Ex. no: 10 b
Date:

Wireless Internetworking

Aim:

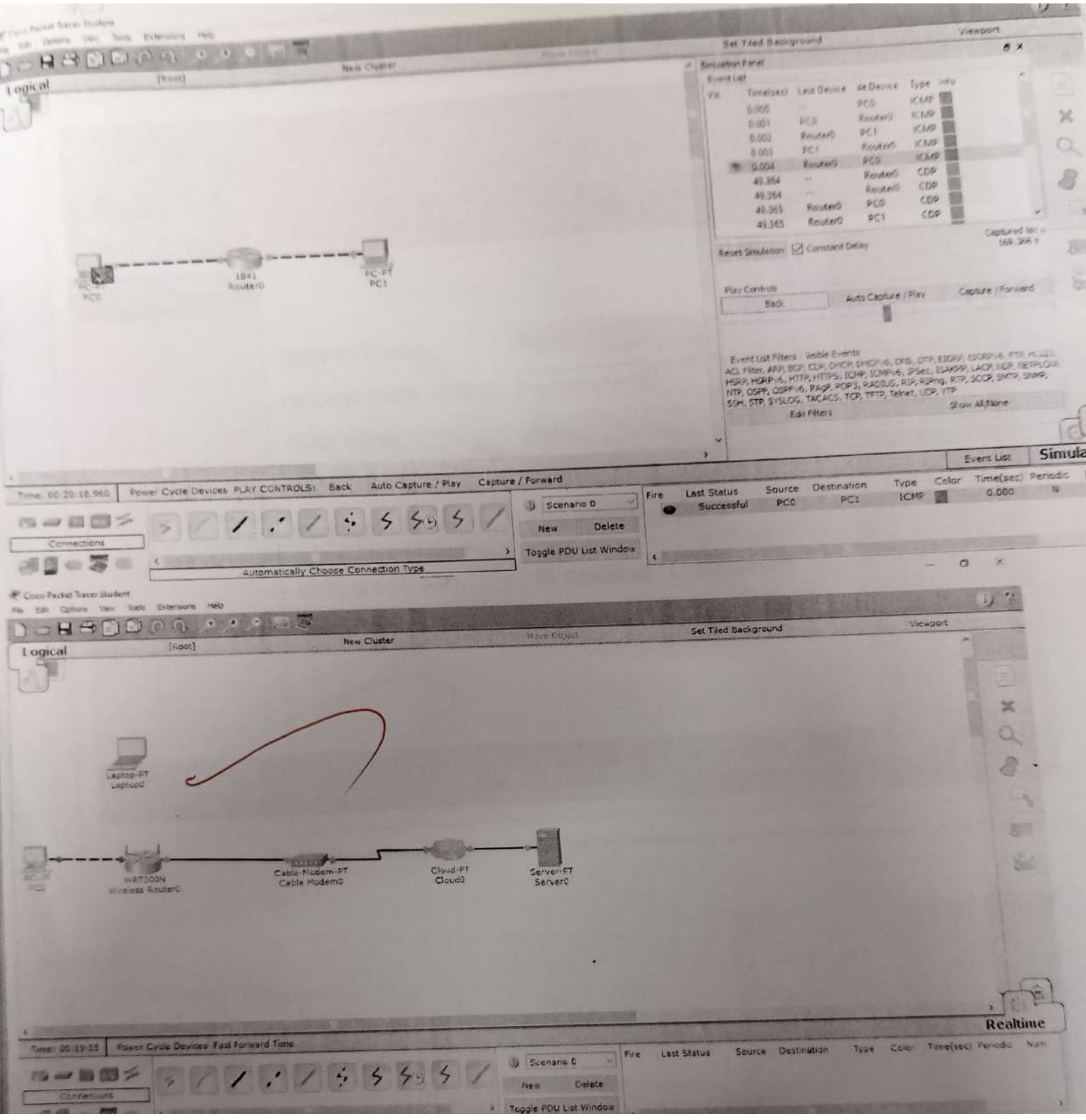
To design and configure an internetwork using a router in Cisco Packet Tracer.

Algorithm:

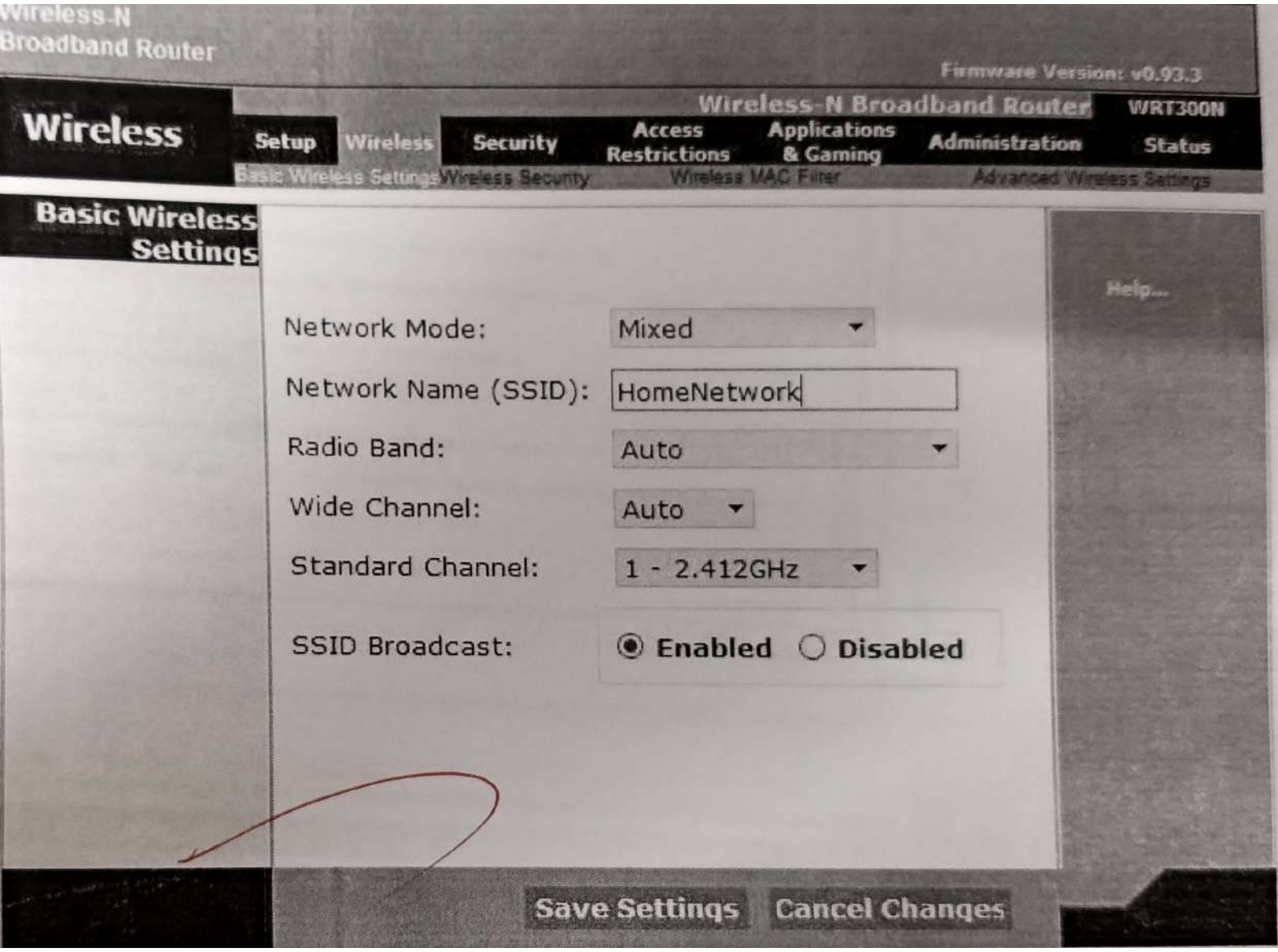
1. Open Cisco Packet Tracer.
2. Add one router and two PCs.
3. Connect the PCs to the router.
4. Configure the router interfaces with IP addresses.
5. Configure the wireless router settings.
6. Configure the DHCP server with IP settings.
7. Enable DHCP for automatic IP assignment.
8. Connect the wireless devices to the network.
9. Verify the assigned IP addresses.
10. Test connectivity b/w the devices and Internet cloud.

Result:

Thus, the wireless network was successfully configured and connectivity was verified.


Logical
[Root]
New Cluster
Set Tiled Background
Viewport
Simulation Panel
Event List
Reset Simulation
Constant Delay
Play Controls
Back
Auto Capture / Play
Capture / Forward
Event List Filters - Visible Events
Edit Filters
Show All/None
1841
Router0
PC-PT
PC1
PC0
Time: 00:20:18.960
Power Cycle Devices
PLAY CONTROLS:
Back
Auto Capture / Play
Capture / Forward
Connections
Automatically Choose Connection Type
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Successful
PC0
PC1
ICMP
0.000
Event List
Simula

Cisco Packet Tracer Student
File Edit Options View Tools Extensions Help
Logical
[Root]
New Cluster
Move Object
Set Tiled Background
Viewport
Laptop-PT
Laptop0
WRT300N
Wireless Router0
Cable-Modem-PT
Cable Modem0
Cloud-PT
Cloud0
Server-PT
Server0
Time: 00:19:55
Power Cycle Devices
Fast Forward Time
Connections
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Realtime


Wireless-N
Broadband Router
Firmware Version: v0.93.3
Wireless
Wireless-N Broadband Router
WRT300N
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Administration
Status
Basic Wireless Settings
Wireless Security
Wireless MAC Filter
Advanced Wireless Settings
Basic Wireless Settings
Help...
Network Mode:
Mixed
Network Name (SSID):
HomeNetwork
Radio Band:
Auto
Wide Channel:
Auto
Standard Channel:
1 - 2.412GHz
SSID Broadcast:
Enabled
Disabled
Save Settings
Cancel Changes

Config CLI

IOS Command Line Interface

```
Press RETURN to get started!


Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to
up
interface FastEthernet0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to
up

Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

PC0

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

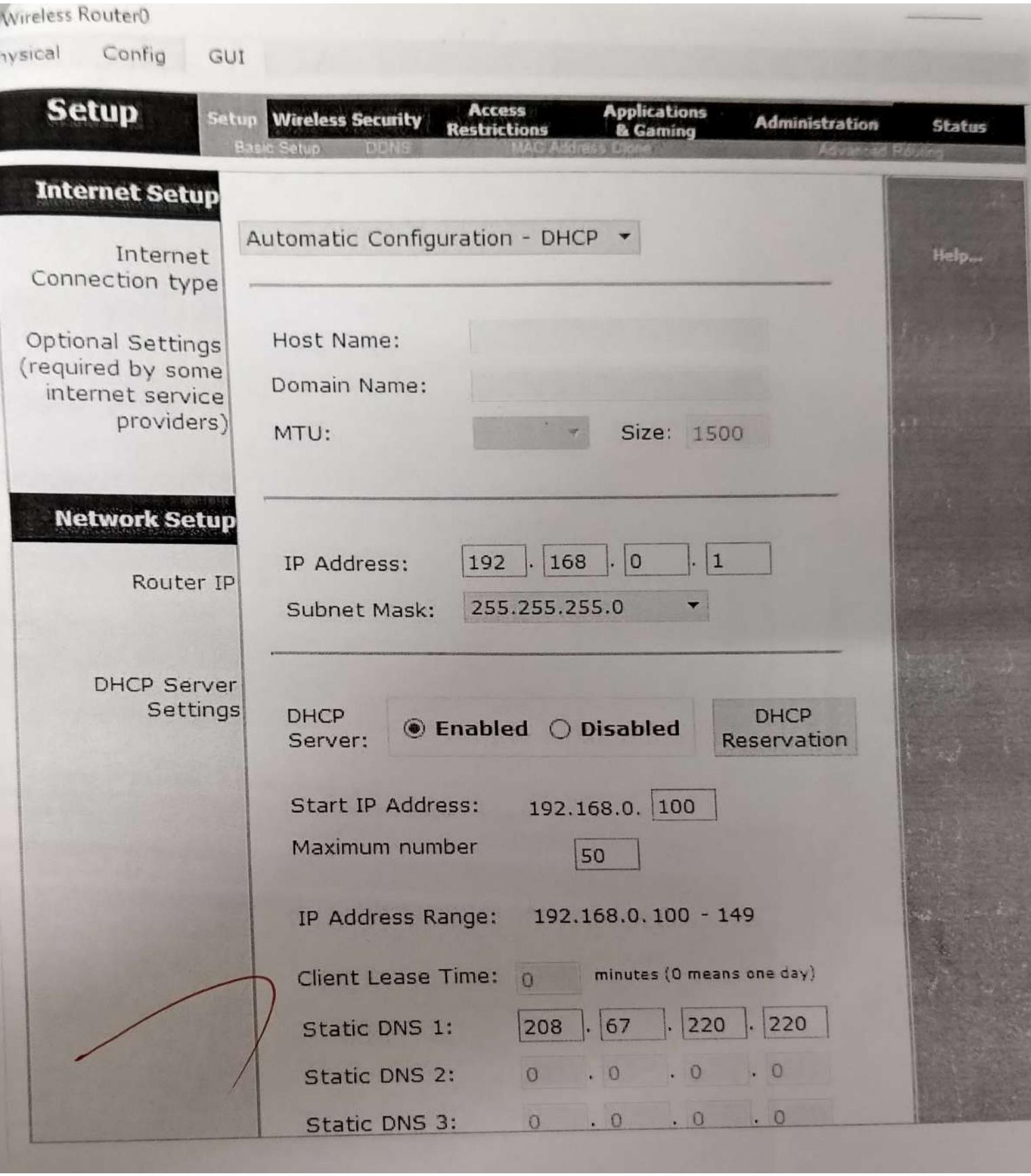
- ○ DHCP
- ◉ Static

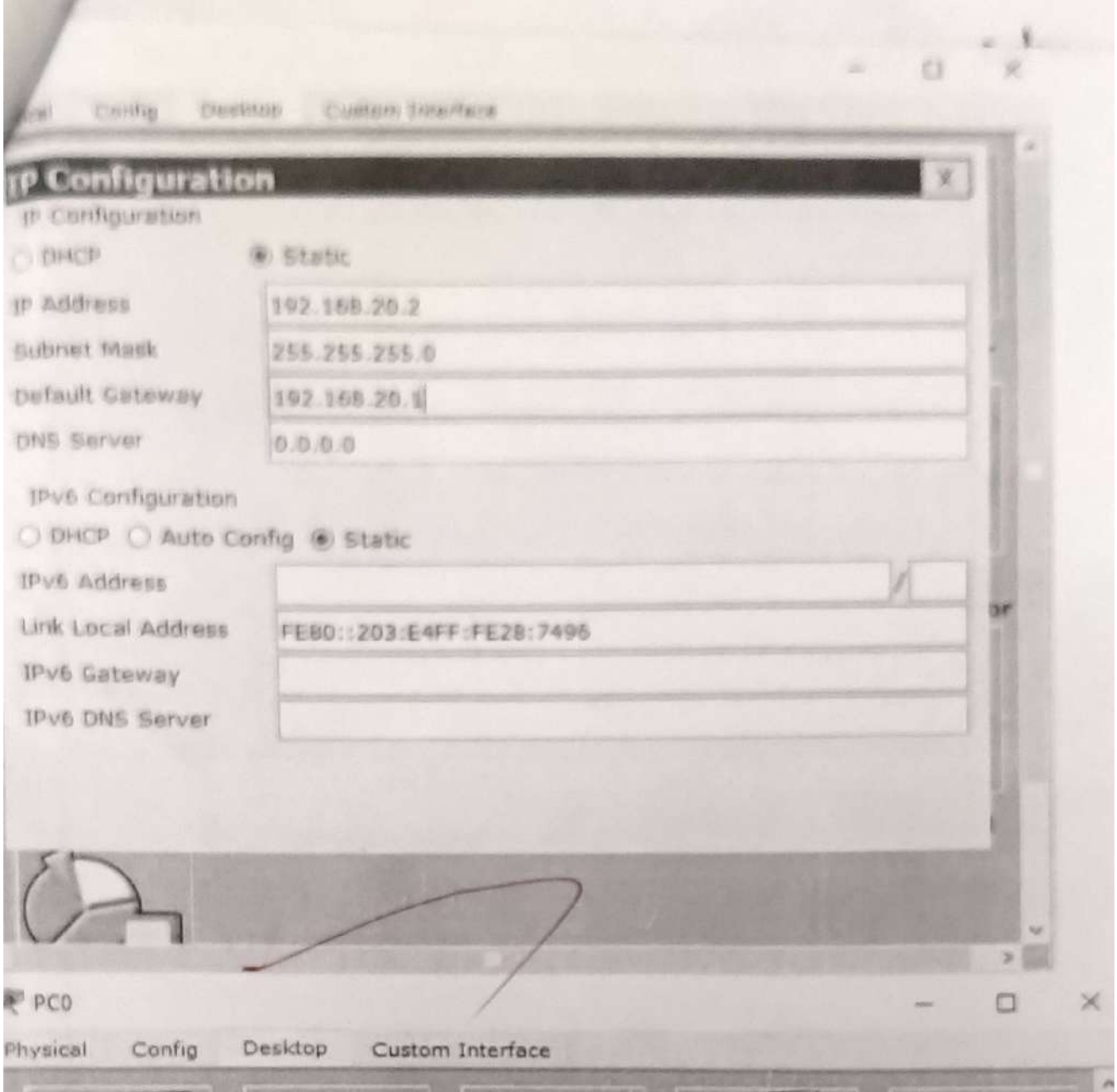
IP Address	192.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	

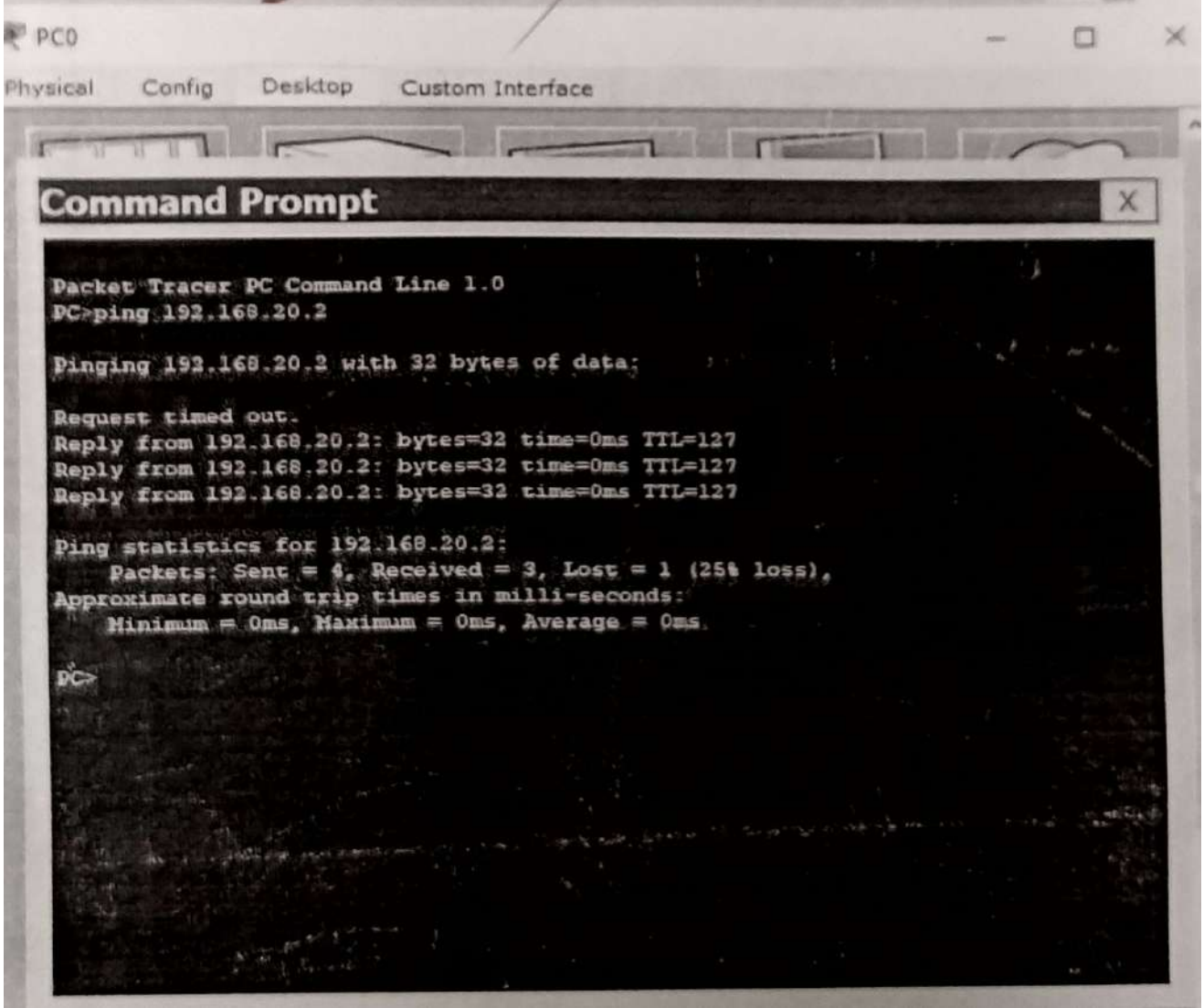
IPv6 Configuration

- ○ DHCP
- ○ Auto Config
- ◉ Static

IPv6 Address	/
Link Local Address	FE80::201:42FF:FE97:C1DE
IPv6 Gateway	
IPv6 DNS Server	


Wireless Router0
Physical
Config
GUI
Setup
Setup
Wireless Security
Access Restrictions
Applications & Gaming
Administration
Status
Basic Setup
DDNS
MAC Address Clone
Advanced Routing
Internet Setup
Internet Connection type
Automatic Configuration - DHCP
Help...
Optional Settings (required by some internet service providers)
Host Name:
Domain Name:
MTU:
Size: 1500
Network Setup
Router IP
IP Address: 192 . 168 . 0 . 1
Subnet Mask: 255.255.255.0
DHCP Server Settings
DHCP Server: Enabled Disabled
DHCP Reservation
Start IP Address: 192.168.0. 100
Maximum number 50
IP Address Range: 192.168.0.100 - 149
Client Lease Time: 0 minutes (0 means one day)
Static DNS 1: 208 . 67 . 220 . 220
Static DNS 2: 0 . 0 . 0 . 0
Static DNS 3: 0 . 0 . 0 . 0


Config
Desktop
Custom Interface
IP Configuration
IP Configuration
DHCP
Static
IP Address
192.168.20.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::203:E4FF:FE28:7496
IPv6 Gateway
IPv6 DNS Server


PC0
Physical
Config
Desktop
Custom Interface
Command Prompt
Packet Tracer PC Command Line 1.0
PC>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.20.2: bytes=32 time=0ms TTL=127
Reply from 192.168.20.2: bytes=32 time=0ms TTL=127
Reply from 192.168.20.2: bytes=32 time=0ms TTL=127
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
PC>

Ex. no: 10a
Date:

Internetworking with Router

Aim:

To design and configure an internetwork using a router in Cisco Packet Tracer.

Algorithm:

1. Open Cisco Packet Tracer.
2. Add one router and two PCs.
3. Connect the PCs to the router.
4. Configure the router interfaces with IP addresses.
5. Enable the interfaces using no shutdown.
6. Configure IP address and subnet mask for PC0.
7. Configure IP address and subnet mask for PC1
8. Set the default gateways for both PCs.
9. Send a PDU from PC0 to PC1
10. Verify the communication b/w two networks.

Ex. no: 10a
Date:

Internetworking with Router

Aim:

To design and configure an internetwork using a router in Cisco Packet Tracer.

Algorithm:

1. Open Cisco Packet Tracer.
2. Add one router and two PCs.
3. Connect the PCs to the router.
4. Configure the router interfaces with IP addresses.
5. Enable the interfaces using no shutdown.
6. Configure IP address and subnet mask for PC0.
7. Configure IP address and subnet mask for PC1
8. Set the default gateways for both PCs.
9. Send a PDU from PC0 to PC1
10. Verify the communication b/w two networks.

Ex no: 11 a

Static Routing

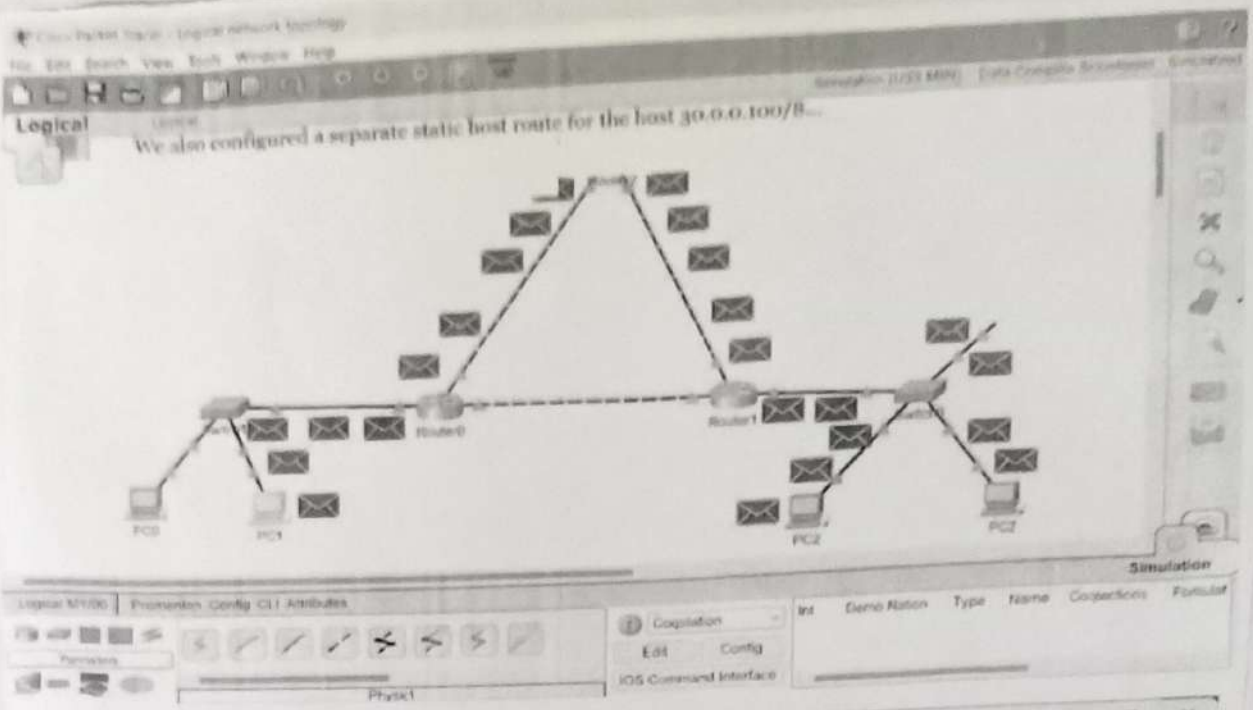
Date:

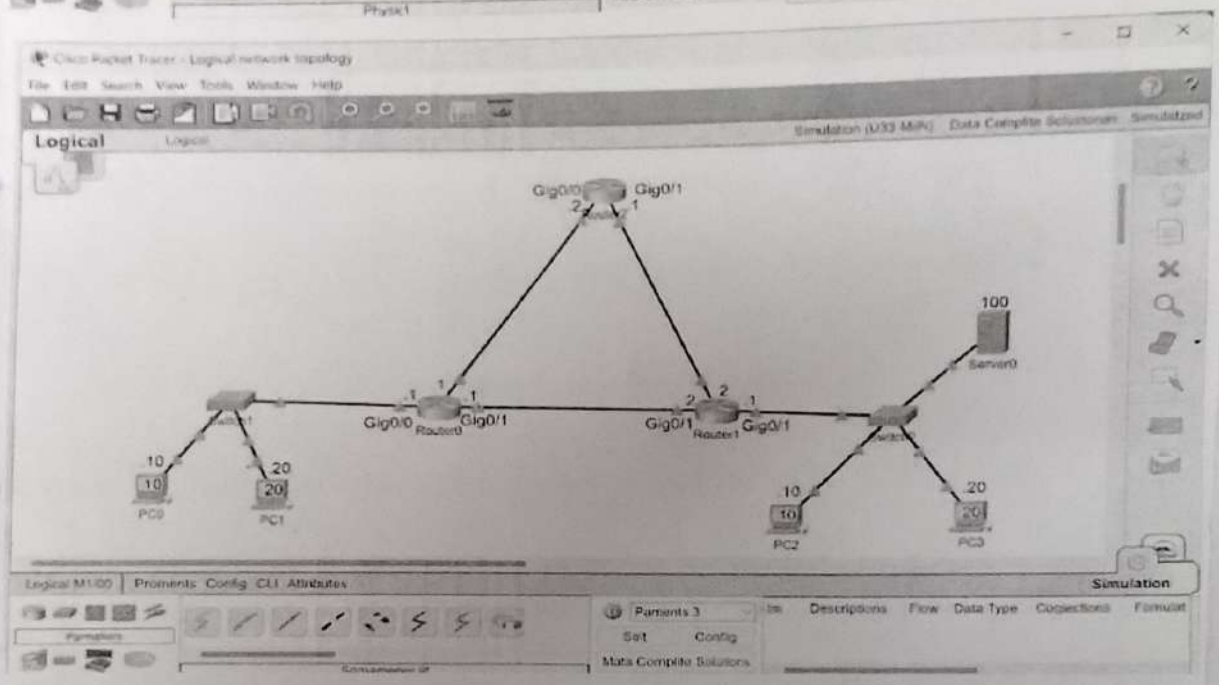
Aim:

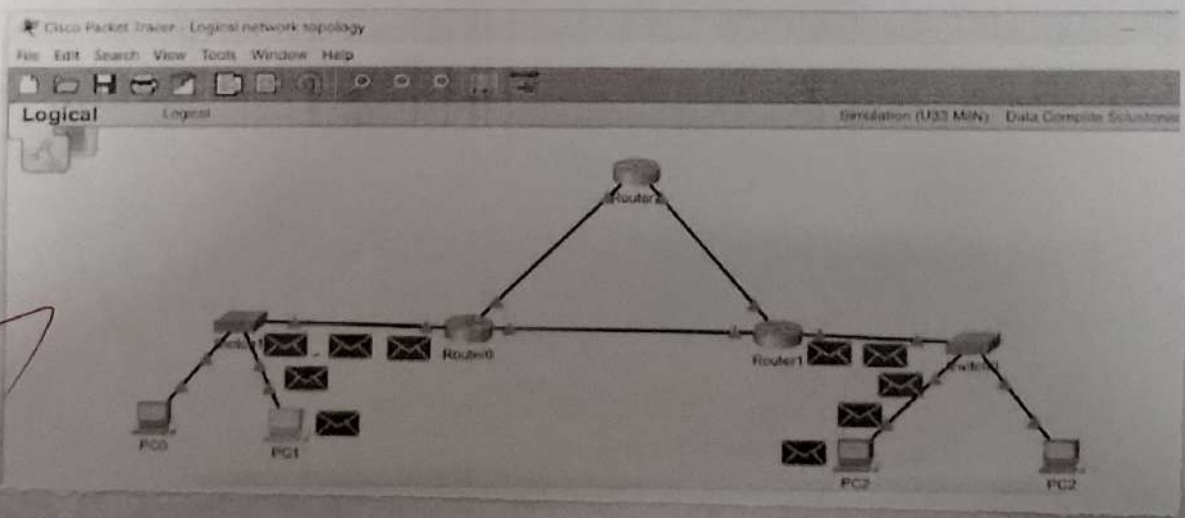
To configure static routing using Cisco Packet Tracer.

Algorithm:

1. Open Cisco Packet Tracer.
2. Create a required router and PC topology
3. Connect the devices using suitable cables
4. Assign IP addresses to PCs and router interfaces
5. Enable router interfaces using no shutdown.
6. Configure static routes using ip route.
7. Configure the required backup route.
8. Check the routing table using show ip route.
9. Test connectivity using ping.
10. Verify communication b/w the networks.







Result:

Thus, static routing was successfully configured and connectivity was verified.

Ex. no: 11b RIP Routing

Date:

Aim:

To configure RIP routing using Cisco Packet Tracer.

Algorithm:

1. Open Cisco Packet Tracer.
2. Create the required routers and PC topology.
3. Connect the devices using suitable cables.
4. Assign IP addresses to PCs and router interfaces.
5. Enable router interfaces using no shutdown.
6. Enter RIP configuration mode using router rip.
7. Add the connected networks using network commands.
8. Configure RIP on the required routers.
9. Check the routing table.
10. Test connectivity using ping.

Result:

Thus, RIP routing was successfully configured and connectivity b/w the networks was verified.

Ex. no: 12a Echo Client-Server

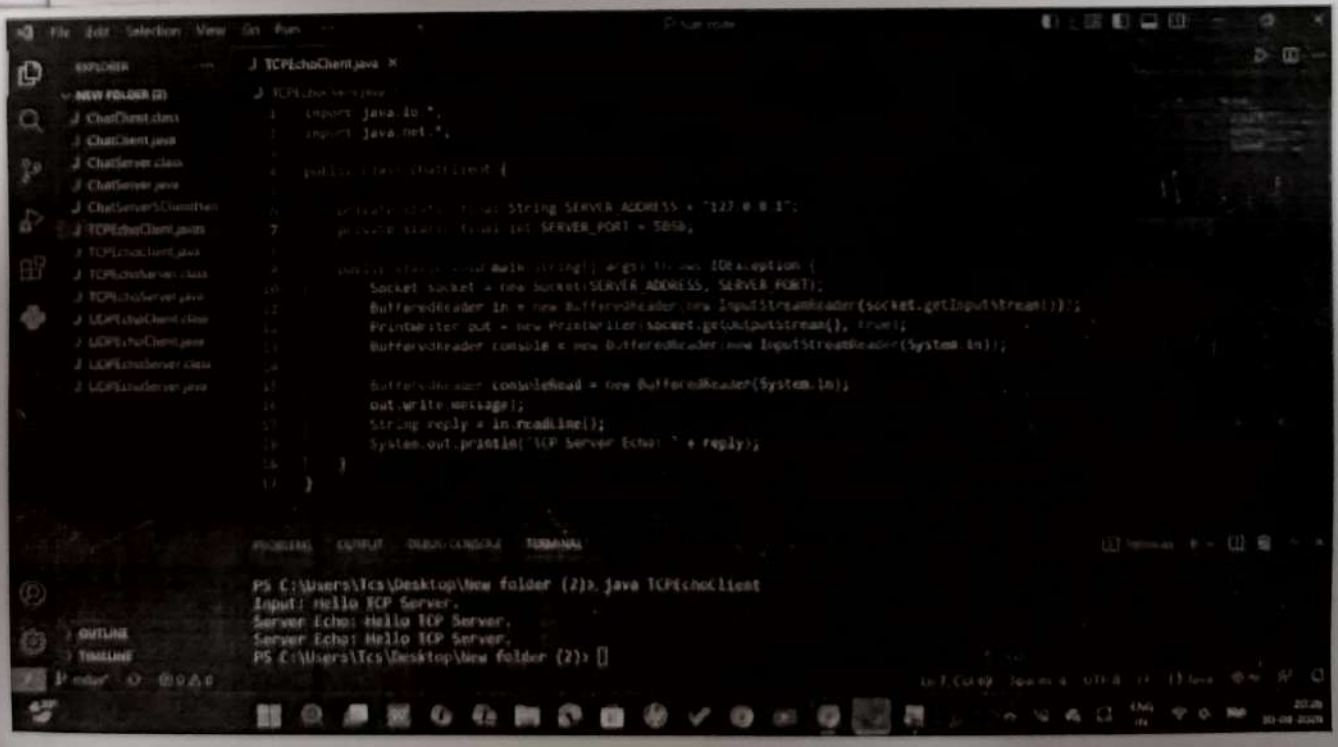
Date:

Aim:

To implement an echo client-server using TCP/UDP sockets.

Algorithm:

1. Create the server socket.
2. Bind the socket to an IP address and port.
3. Start listening for client requests.
4. Create the client socket.
5. Connect the client to the server.
6. Enter a message from the client.
7. Send the message to the server.
8. Server receives and echoes the message.
9. Client receives the echoed message.
10. Close the sockets.



Ex. no: 12a Echo Client-Server

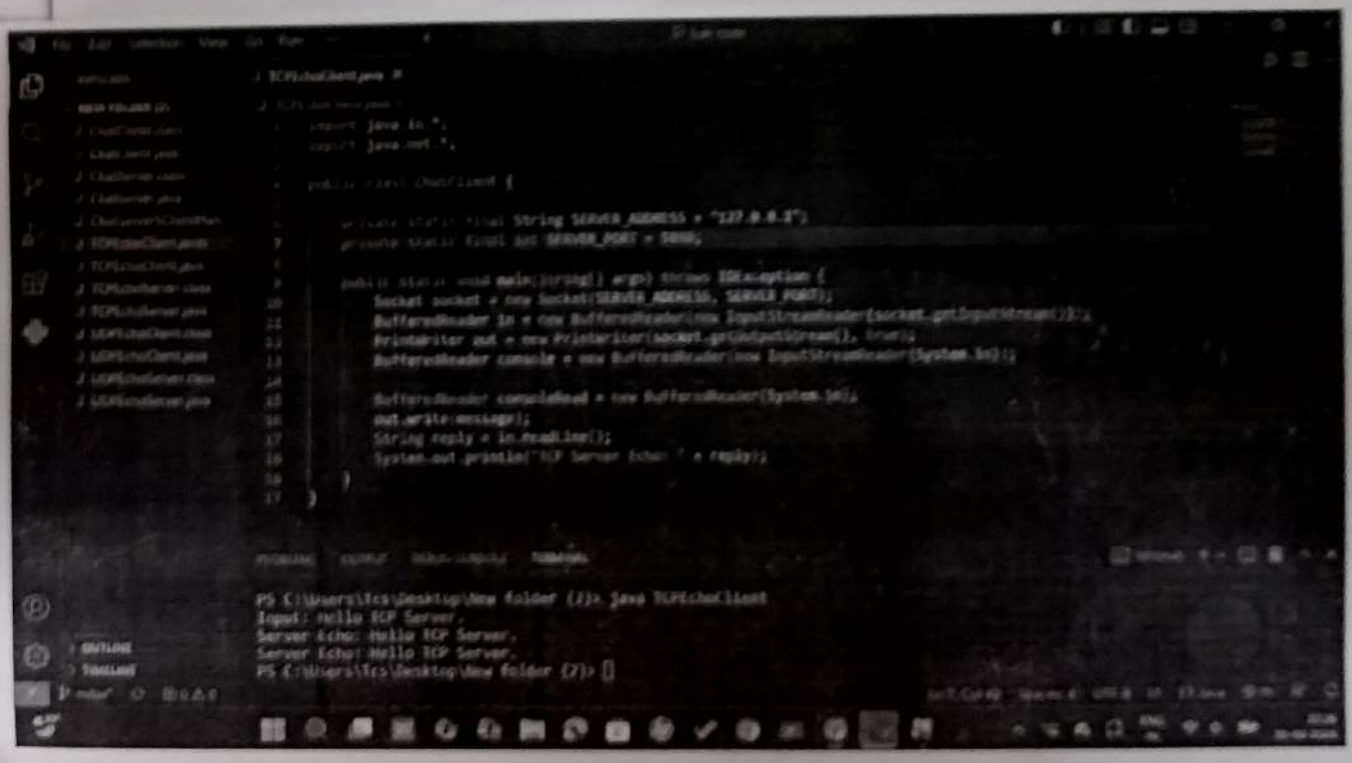
Date:

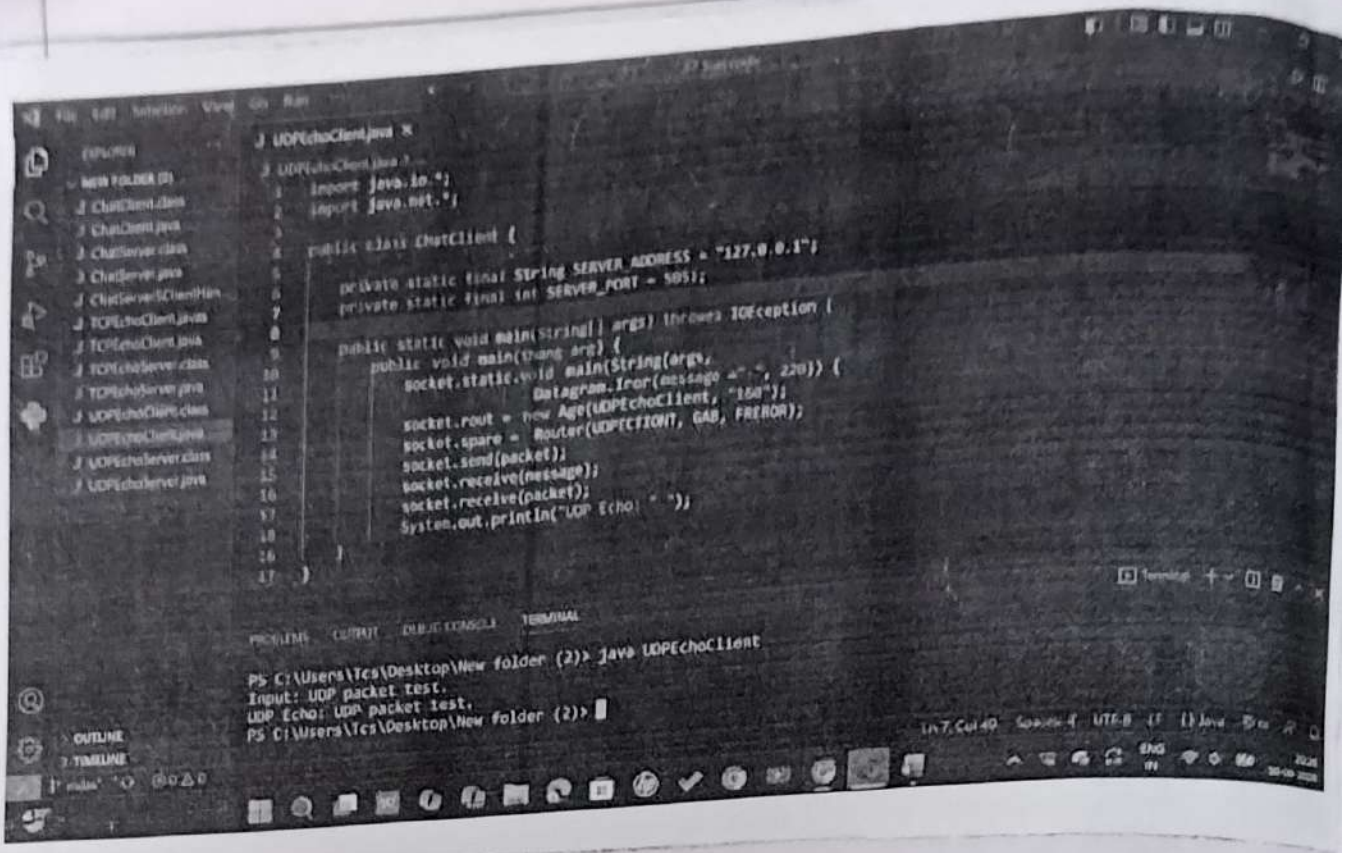
Aim:

To implement an echo client-server using TCP/UDP sockets

Algorithm:

1. Create the server socket
2. Bind the socket to an IP address and port.
3. Start listening for client requests.
4. Create the client socket.
5. Connect the client to the server.
6. Enter a message from the client.
7. Send the message to the server
8. Server receives and echoes the message.
9. Client receives the echoed message
10. Close the sockets.





Result:

Thus, the echo client - server was successfully implemented using sockets.

Ex.no: 10b Chat - Client - server

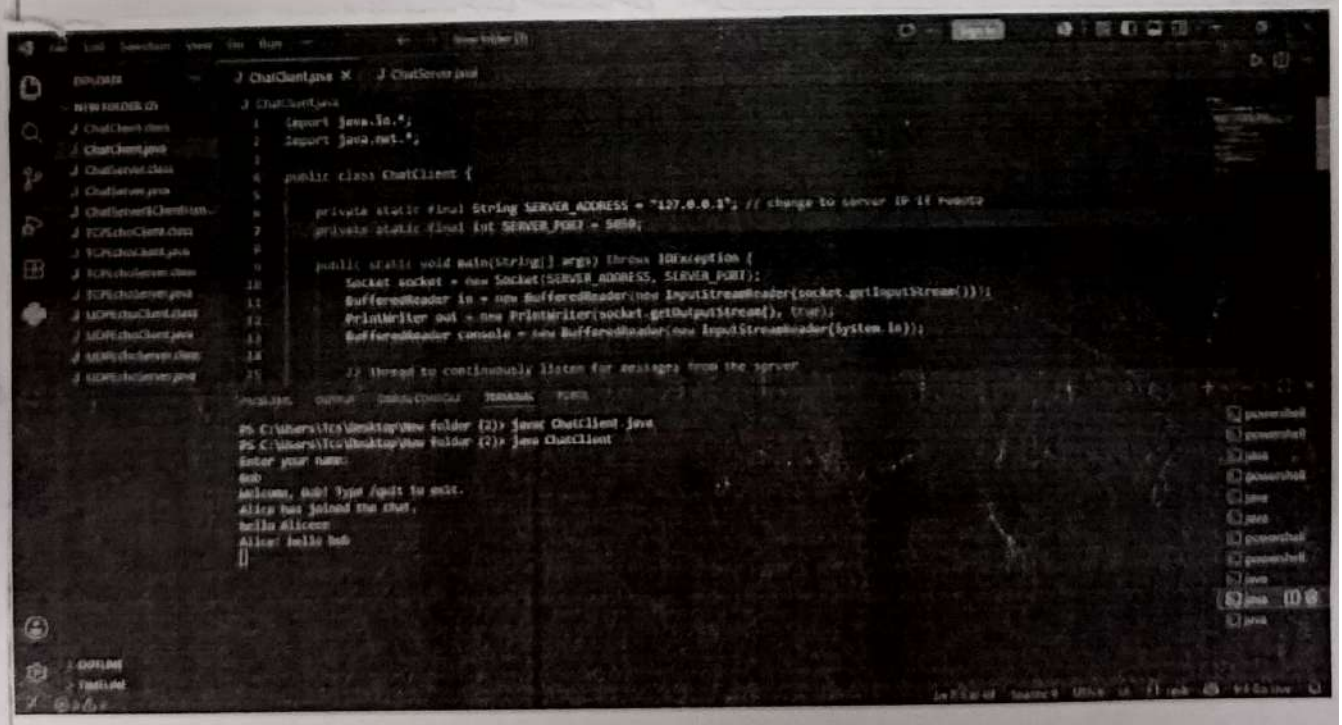
Date:

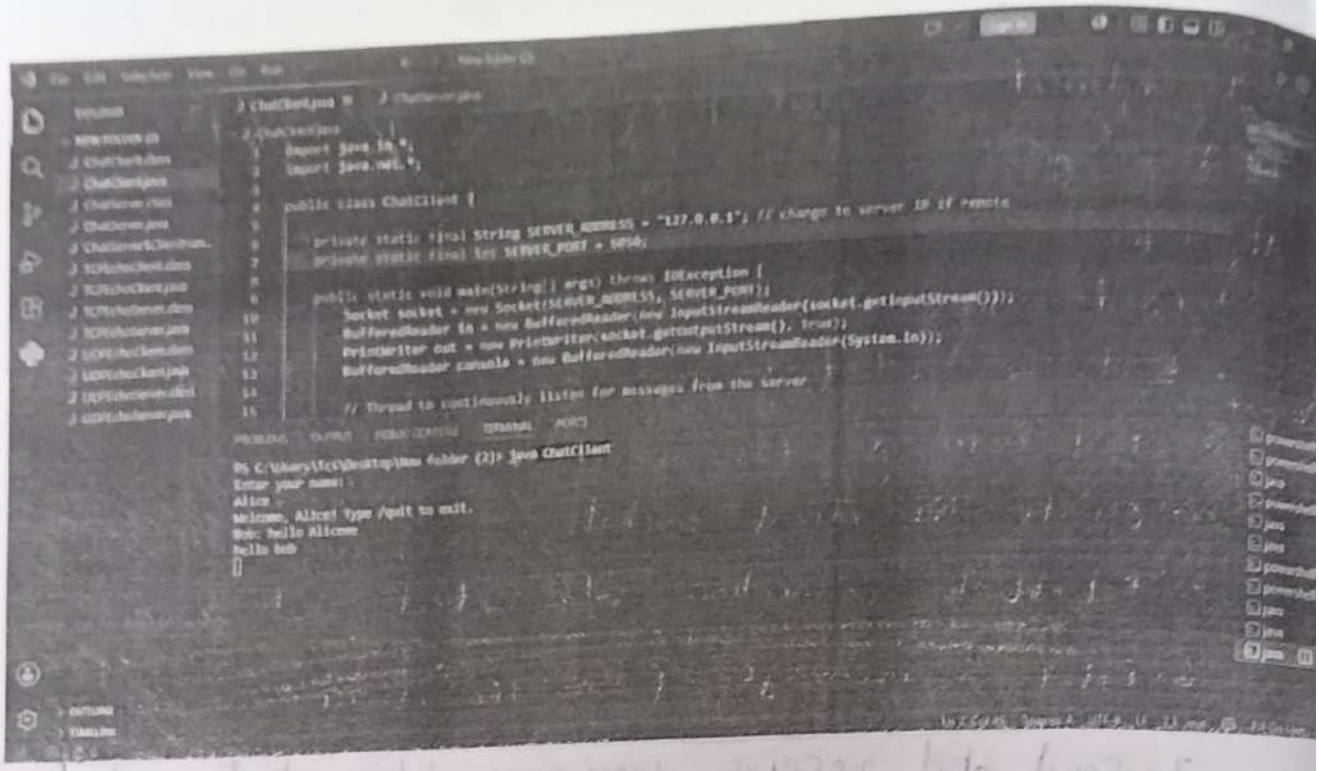
Aim:

To implement a chat program using socket programming.

Algorithm:

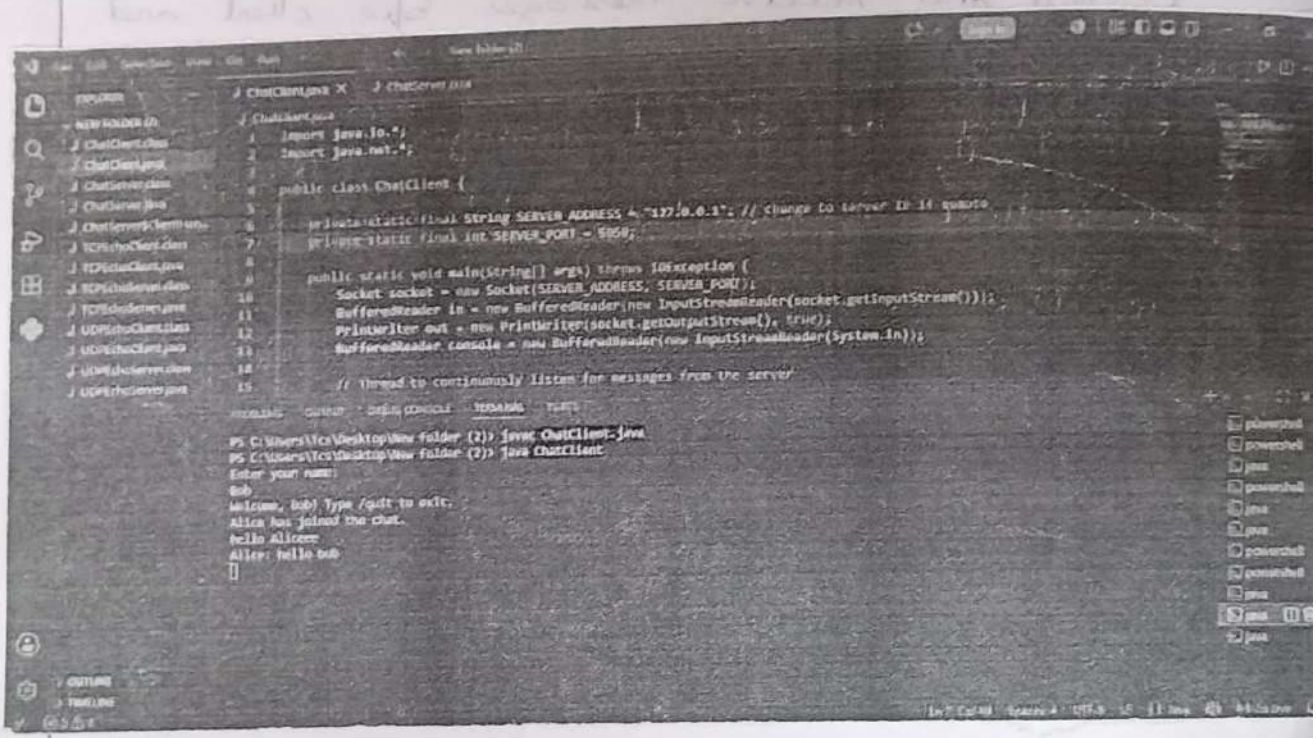
1. Create the server socket
2. Bind the socket to an IP address and port.
3. Start listinerg for a client
4. Create the client socket.
5. Establish connection b/w client and server
6. Enter a message from the client.
7. Send and receive messages b/w client and server.
8. Display the received messages.
9. Continue communication until termination.
10. Close the sockets.





```
ChatClient.java
1  import java.io.*;
2  import java.net.*;
3
4  public class ChatClient {
5
6      private static final String SERVER_ADDRESS = "127.0.0.1"; // change to server IP if remote
7      private static final int SERVER_PORT = 5050;
8
9      public static void main(String[] args) throws IOException {
10         Socket socket = new Socket(SERVER_ADDRESS, SERVER_PORT);
11         BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
12         PrintWriter out = new PrintWriter(socket.getOutputStream(), true);
13         BufferedReader console = new BufferedReader(new InputStreamReader(System.in));
14
15         // Thread to continuously listen for messages from the server
```

```
PS C:\Users\Tcs\Desktop\New folder (2)> java ChatClient
Enter your name:
Alice
Welcome, Alice! Type /quit to exit.
Bob: hello Alicee
hello bob
```



```
ChatClient.java
1  import java.io.*;
2  import java.net.*;
3
4  public class ChatClient {
5
6      private static final String SERVER_ADDRESS = "127.0.0.1"; // change to server IP if remote
7      private static final int SERVER_PORT = 5050;
8
9      public static void main(String[] args) throws IOException {
10         Socket socket = new Socket(SERVER_ADDRESS, SERVER_PORT);
11         BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
12         PrintWriter out = new PrintWriter(socket.getOutputStream(), true);
13         BufferedReader console = new BufferedReader(new InputStreamReader(System.in));
14
15         // Thread to continuously listen for messages from the server
```

```
PS C:\Users\Tcs\Desktop\New folder (2)> javac ChatClient.java
PS C:\Users\Tcs\Desktop\New folder (2)> java ChatClient
Enter your name:
Bob
Welcome, Bob! Type /quit to exit.
Alice has joined the chat.
hello Alicee
Alice: hello bob
```

Result:

Thus, the chat client-server was successfully implemented using socket programming.

Ex. no: 13 Implement your own Ping Program
Date:

Aim:
To Implement a Ping program using python raw sockets to check the reachability of a host and calculate the round-trip time.

Python Program:

```
import socket
import struct
import time
import os

def checksum(data):
    total = 0
    for i in range(0, len(data) - 1, 2):
        total += (data[i] << 8) + data[i+1]
    if len(data) % 2:
        total += data[-1] << 8
    total = (total >> 16) + (total & 0xffff)
    total += total >> 16
    return (~total) & 0xffff

def ping(host):
    try:
        dest_addr = socket.gethostbyname(host)
        sock = socket.socket(socket.AF_INET, socket.SOCK_RAW,
                             socket.IPPROTO_ICMP)

        packet_id = os.getpid() & 0xffff
        sequence = 1
        header = struct.pack("!BBHHH", 8, 0, 0, packet_id, sequence)
        data = b"PING"
        chksum = checksum(header + data)
        header = struct.pack("!BBHHH", 8, 0, chksum, packet_id,
                             sequence)
        packet = header + data
        start = time.time()
```

```
            sock.sendto(packet,(dest_addr,1))
            sock.settimeout(2)
            sock.recvfrom(1024)
            end = time.time()
            att = (end - start) * 1000
            print "Reply from", dest_addr, ":time = %.2f ms" %.n
            sock.close()
        except socket.timeout:
            print("Request timed out")
        except PermissionError:
            print("Permission denied. Run the program as admi
                  root.")
host = input("Enter destination host:")
ping(host)
```

Sample Input:

Enter destination host: 8.8.8.8

Sample Output:

Reply from 8.8.8.8 : time = 24.53 ms

Result:

Thus, the ping program was successfully implem
using python.

Ex. no: 14 Packet Sniffing Using Raw Sockets

Date:

Aim:

To implement a packet sniffing program using Python raw sockets and display the source IP, destination IP and protocol of captured packets.

Code:

```
import socket
import struct
sock = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.
                                                IPPROTO_IP)

print("Packet Sniffer Started...")
print("Waiting for packets...\n")
while True:
    packet, address = sock.recvfrom(65535)
    ip_header = packet[0:20]
    iph = struct.unpack("!BBHHHBBH4s4s", ip_header)
    source_ip = socket.inet_ntoa(iph[8])
    destination_ip = socket.inet_ntoa(iph[9])
    protocol = iph[6]
    print("Source IP : ", source_ip)
    print("Destination IP: ", destination_ip)
    print("Protocol : ", protocol)
    print("Packet Size : ", len(packet), 'bytes')
```

Sample Output:

```
Packet Sniffer Started...
Waiting for packets...
Source IP        : 192.168.1.10
Destination IP   :    142.250.183.14
Protocol         :    6
Packet Size      :    60 bytes
```

Source IP : 8.8.8.8
Destination IP : 192.168.1.10
Protocol : 1
Packet Size : 84 bytes

Result:

Thus, the ping program and packet sniffing program were successfully implemented using python raw sockets

Ex.no: 15 Analysis of various types of
Date: servers using webalizer

Aim:

To analyze the web server log files using the webalizer tool and study various server statistics such as number of requests, visitors, transferred data, URLs and response status codes.

Algorithm:

1. Start the webalizer application.
2. Install and configure Webalizer on the system.
3. Locate the web server access log file.
4. Configure Webalizer with the required log-file path and output directory.
5. Run Webalizer on the selected log file.
6. Webalizer reads and analyzes the server log entries.
7. It generates statistics such as:
   - Number of visits
   - Number of requests
   - Pages accessed
   - Data transfered
   - HTTP response codes
   - Referring sites
   - Operating systems and browsers.
8. Open the generated HTML report in a web browser.
9. Analyze the server activity using generated graphs and tables.
10. Repeat the analysis for different server log files if required.
11. Compare the obtained server statistics.
12. Record the observations and terminate the experiment

Input:

Web server access log files containing information such as:

Client IP
Data and Time
Requested URL
HTTP Method
HTTP Status Code
Bytes Transferred
Referrer
User Agent.

Example Input:

access.log

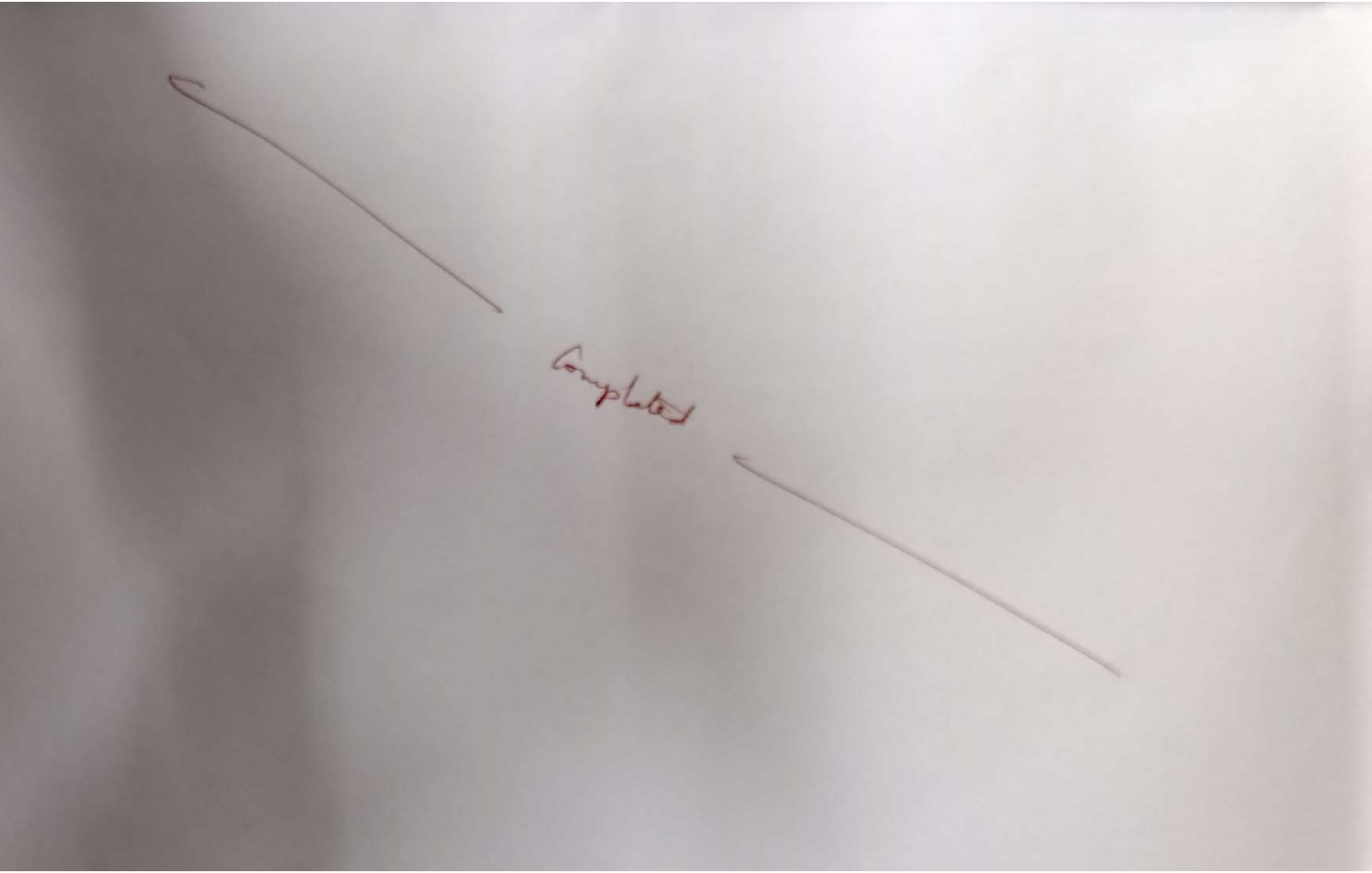
Output:

Webalizer Statistics

Total requests	:	1250
Total visits	:	480
Total pages	:	920
Total bytes	:	15.4 mb
Succesful Requests	:	1180
404 Errors	:	45

Result:

Thus, the webalizer tool was used successfully to analyze web server log files, and various server statistics such as requests, visitors, pages, bandwidth usage and HTTP response codes were obtained.


Completed